WHO AM I?

Creative

Crocheting

Painting murals

Designer

Masters in Design @ University of Edinburgh

Product Designer

Accenture

Microsoft

Google

Yoga Teacher

Have students with chronic conditions

YTT at Folk Studio in NOPA

Critical Problem Solver

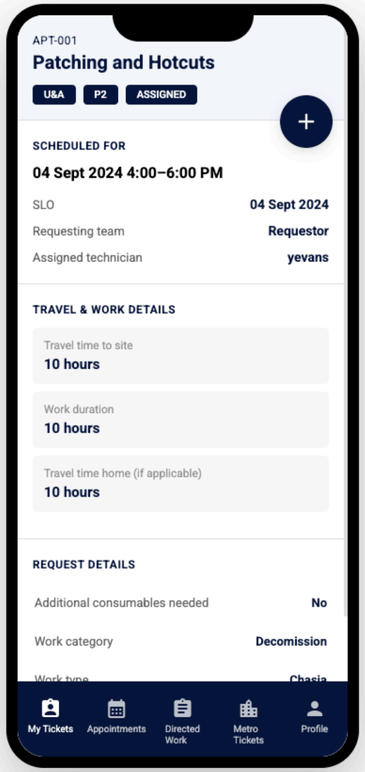
CS @ Northeastern

Front end Engineer @ Rasa (AI startup)

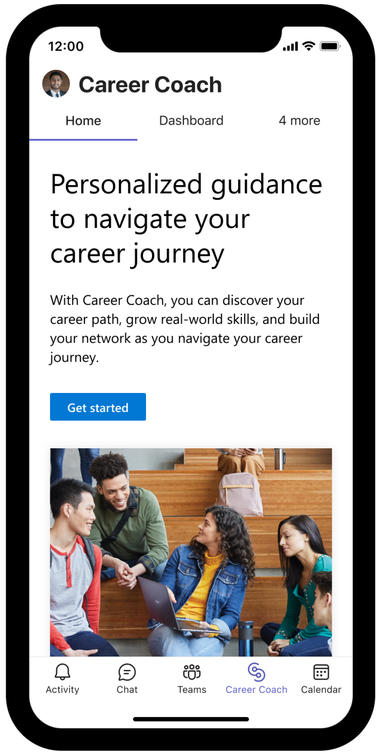
AGENDA

Google NetOps

Moose



Metro Owner platform + field worker mobile app that made automation possible across **10** workstreams, **4** timezones, and **3** user groups. **30%** productivity gain.



Microsoft

Learning Accelerators

Frameworks for progressive disclosure of Insights

Increase Insights Discoverability by **40%**

GOOGLE NET OPS- MEET THE USERS

Field Worker

Metro Owner

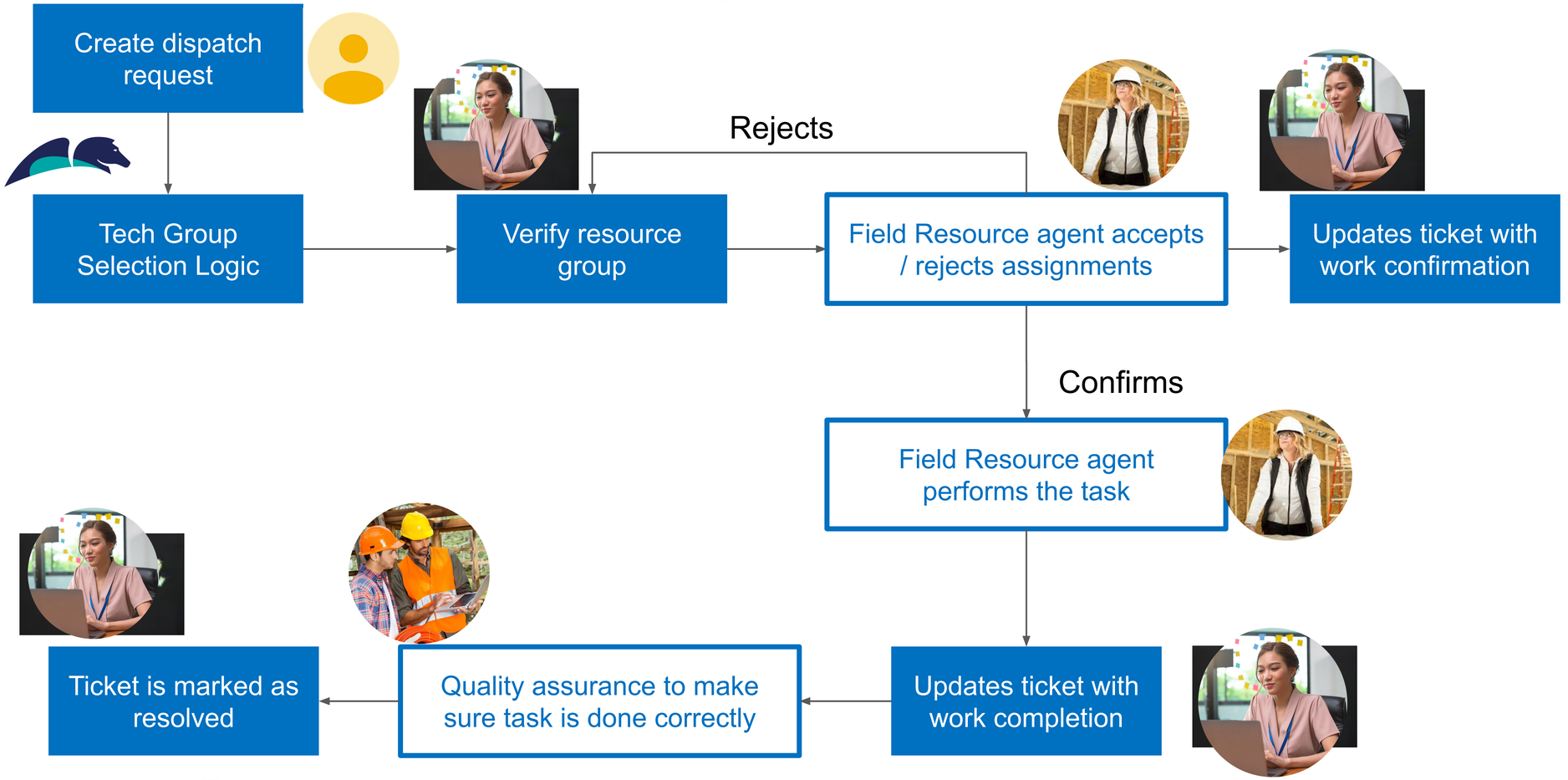
MAD Agent

1. DESIGN ‘THE PROCESS’ FIRST

2. INTERACTION DESIGN

THE MANUAL PROCESS

Decisioning Logic: Today...

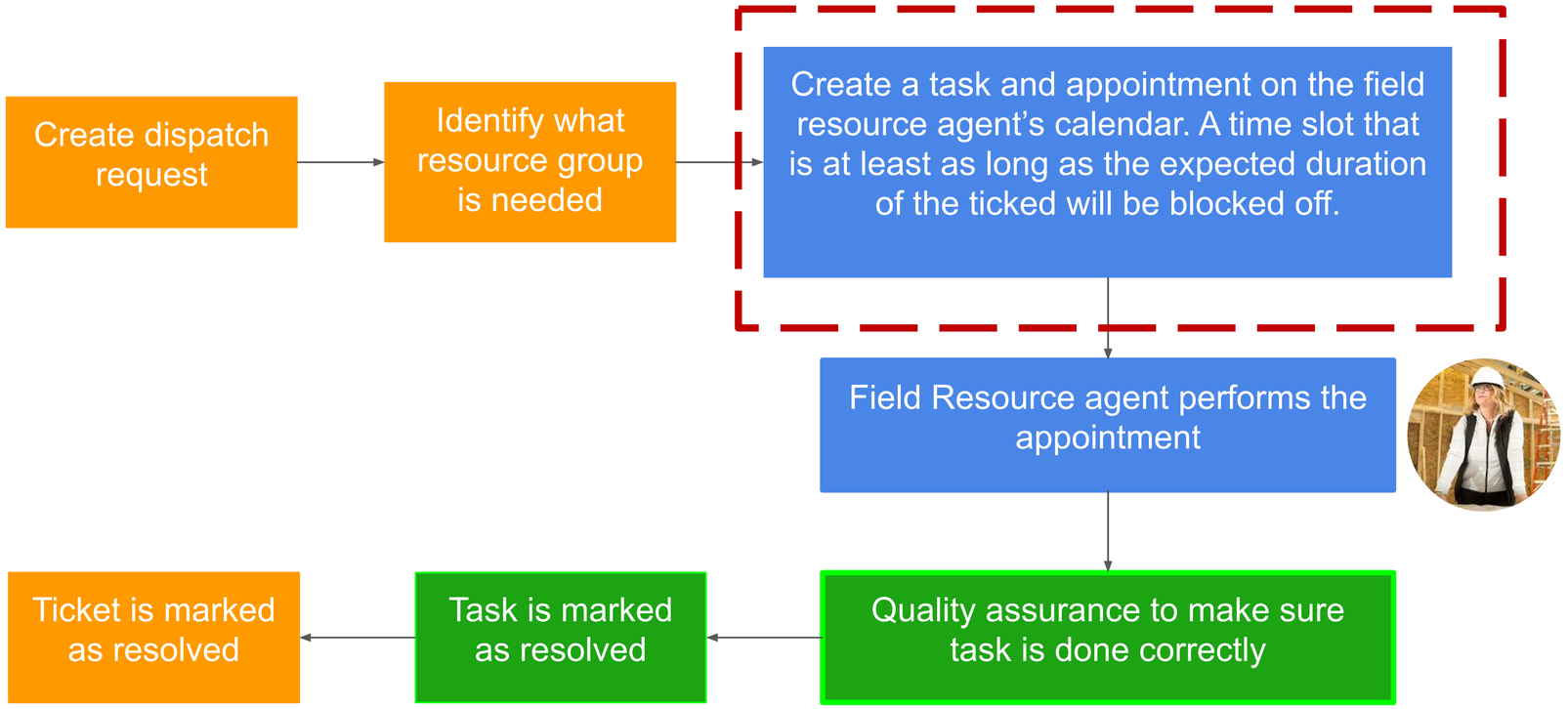


Google | accenture

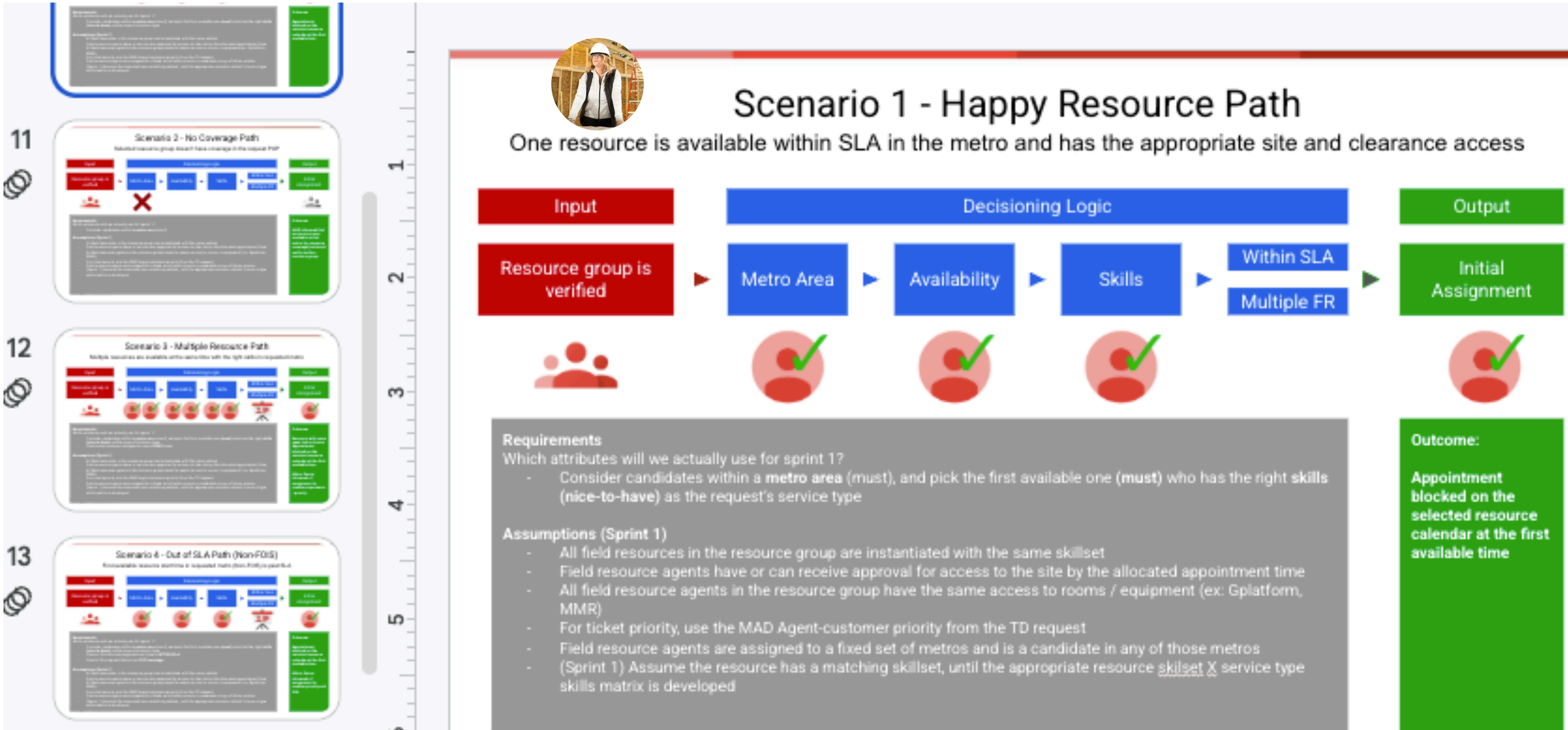
THE BRIEF

"How might we automate the assignment of field workers to tickets — at scale, across regions, in real time?"

Today's Focus: Assignment to a Field Service Agent

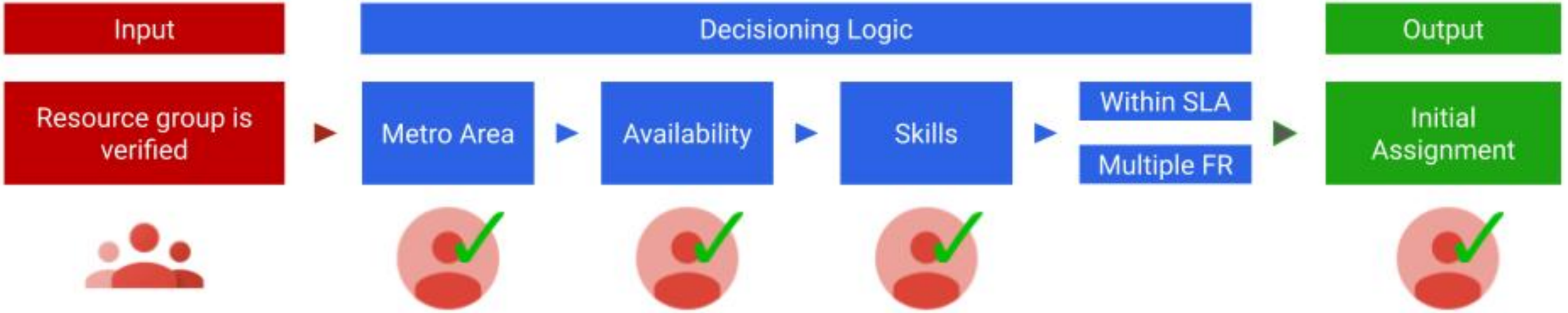


SCREENSHOTS OF SCENARIO MAPPING EXERCISE



Scenario 1 - Happy Resource Path

One resource is available within SLA in the metro and has the appropriate site and clearance access



Requirements

Which attributes will we actually use for sprint 1?

- Consider candidates within a **metro area** (must), and pick the first available one **(must)** who has the right **skills (nice-to-have)** as the request's service type

Assumptions (Sprint 1)

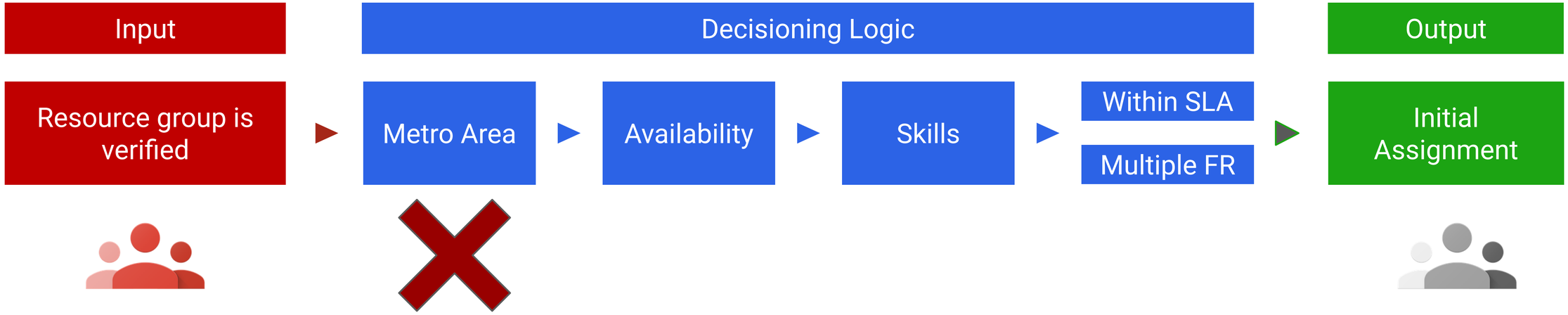
- All field resources in the resource group are instantiated with the same skillset
- Field resource agents have or can receive approval for access to the site by the allocated appointment time
- All field resource agents in the resource group have the same access to rooms / equipment (ex: Gplatform, MMR)
- For ticket priority, use the MAD Agent-customer priority from the TD request
- Field resource agents are assigned to a fixed set of metros and is a candidate in any of those metros
- (Sprint 1) Assume the resource has a matching skillset, until the appropriate resource skilset X service type skills matrix is developed

Outcome:

Appointment blocked on the selected resource calendar at the first available time

Scenario 2 - No Coverage Path

Selected resource group doesn't have coverage in the request POP



Requirements
Which attributes will we actually use for sprint 1?

- Consider candidates within a **metro area** (must)

Assumptions (Sprint 1)

- All field resources in the resource group are instantiated with the same skillset
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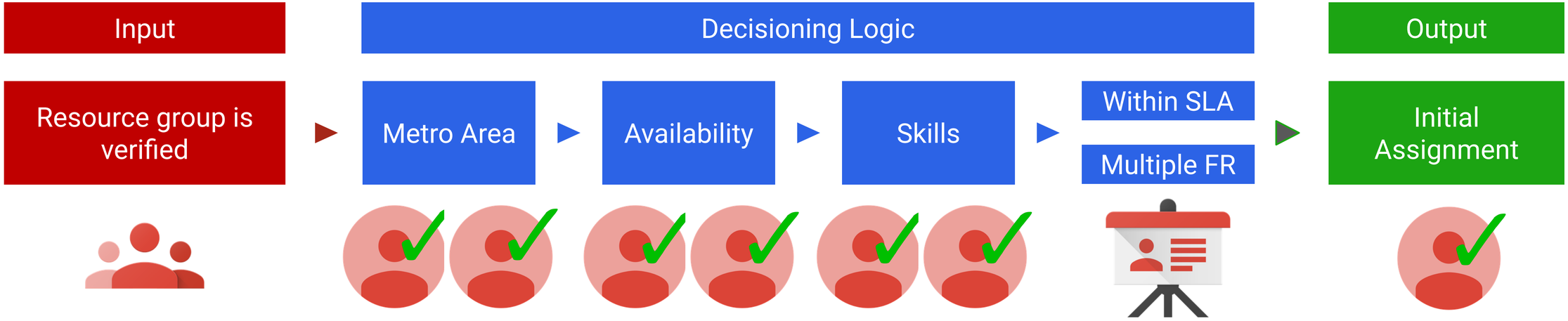
- For ticket priority, use the MAD Agent-customer priority from the TD request
- Field resource agents are assigned to a fixed set of metros and is a candidate in any of those metros
- (Sprint 1) Assume the resource has a matching skillset, until the appropriate resource skilset X service type skills matrix is developed

Outcome:

MAD informed that no resource was available in that metro [no resource coverage] and must verify another resource group

Scenario 3 - Multiple Resource Path

Multiple resources are available at the same time with the right skills in requested metro



Requirements

Which attributes will we actually use for sprint 1?

- Consider candidates within a **metro area** (must), and pick the first available one **(must)** who has the right **skills (nice-to-have)** as the request's service type.
- Check who has been assigned to more **OPEN** tasks.

Assumptions (Sprint 1)

- All field resources in the resource group are instantiated with the same skillset
- Field resource agents have or can receive approval for access to the site by the allocated appointment time
- All field resource agents in the resource group have the same access to rooms / equipment (ex: Gplatform, MMR)
- For ticket priority, use the MAD Agent-customer priority from the TD request
- Field resource agents are assigned to a fixed set of metros and is a candidate in any of those metros
- (Sprint 1) Assume the resource has a matching skillset, until the appropriate resource skilset X service type skills matrix is developed

Outcome:

Resource with more open tasks chosen. Appointment blocked on the selected resource calendar at the first available time.

Metro Owner informed of assignment to confirm experience / priority

Scenario 4 - Out of SLA Path (Non-FOIS)

First available resource start time in requested metro (Non-FOIS) is past SLA

Input	Decisioning Logic				Output
Resource group is verified	Metro Area	Availability	Skills	Within SLA Multiple FR	Initial Assignment

Requirements

Which attributes will we actually use for sprint 1?

- Consider candidates within a **metro area** (must), and pick the first available one **(must)** who has the right **skills (nice-to-have)** as the request's service type.
- Check if the allocated appointment time is **WITHIN SLA.**
- Check if the request Metro has **FOIS coverage**

Assumptions (Sprint 1)

- All field resources in the resource group are instantiated with the same skillset
- Field resource agents have or can receive approval for access to the site by the allocated appointment time
- All field resource agents in the resource group have the same access to rooms / equipment (ex: Gplatform, MMR)
- For ticket priority, use the MAD Agent-customer priority from the TD request
- Field resource agents are assigned to a fixed set of metros and is a candidate in any of those metros
- (Sprint 1) Assume the resource has a matching skillset, until the appropriate resource skilset X service type skills matrix is developed

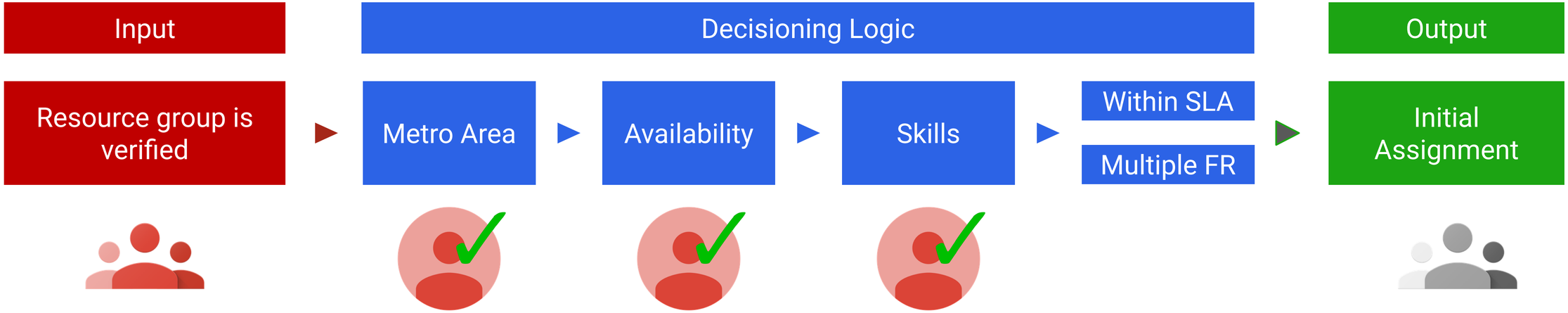
Outcome:

Appointment blocked on the selected resource calendar at the first available time.

Metro Owner informed of assignment to confirm priority and SLA

Scenario 5 - Out of SLA Path (FOIS)

First available resource start time in requested metro (FOIS) is past SLA



Requirements

Which attributes will we actually use for sprint 1?

- Consider candidates within a **metro area** (must), and pick the first available one **(must)** who has the right **skills (nice-to-have)** as the request's service type.
- Check if the allocated appointment time is **WITHIN SLA.**
- Check if the request Metro has **FOIS coverage**

Assumptions (Sprint 1)

- All field resources in the resource group are instantiated with the same skillset
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- (Sprint 1) Assume the resource has a matching skillset, until the appropriate resource skilset X service type skills matrix is developed

Outcome:

MAD informed that no resource was available in that metro [no resource coverage] and must verify another resource group (FOIS?)

QUESTIONS DURING THE PROCESS

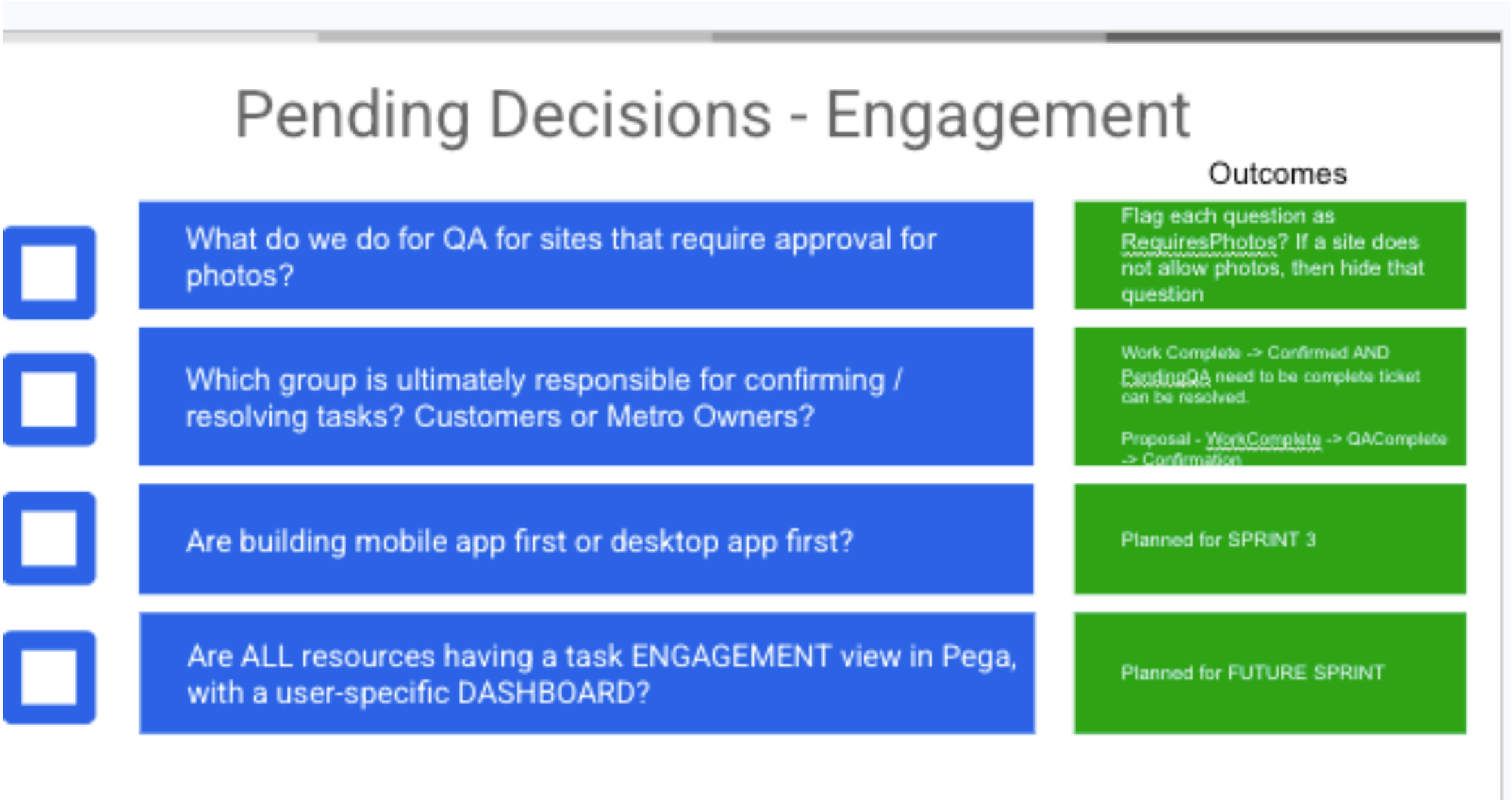
How do we want to assign 8 hour+ estimated durations? Multiple days/ techs?

For Sprint 1, how are we calculating Estimated Travel Duration? Proxy for distance? Before/after buffer?

If a calendared resource is NOT first on the prioritization, how do we engage the resource?

Should we take into account MFS TVS resources that can work overtime?

SCREENSHOTS OF “THE MESS”

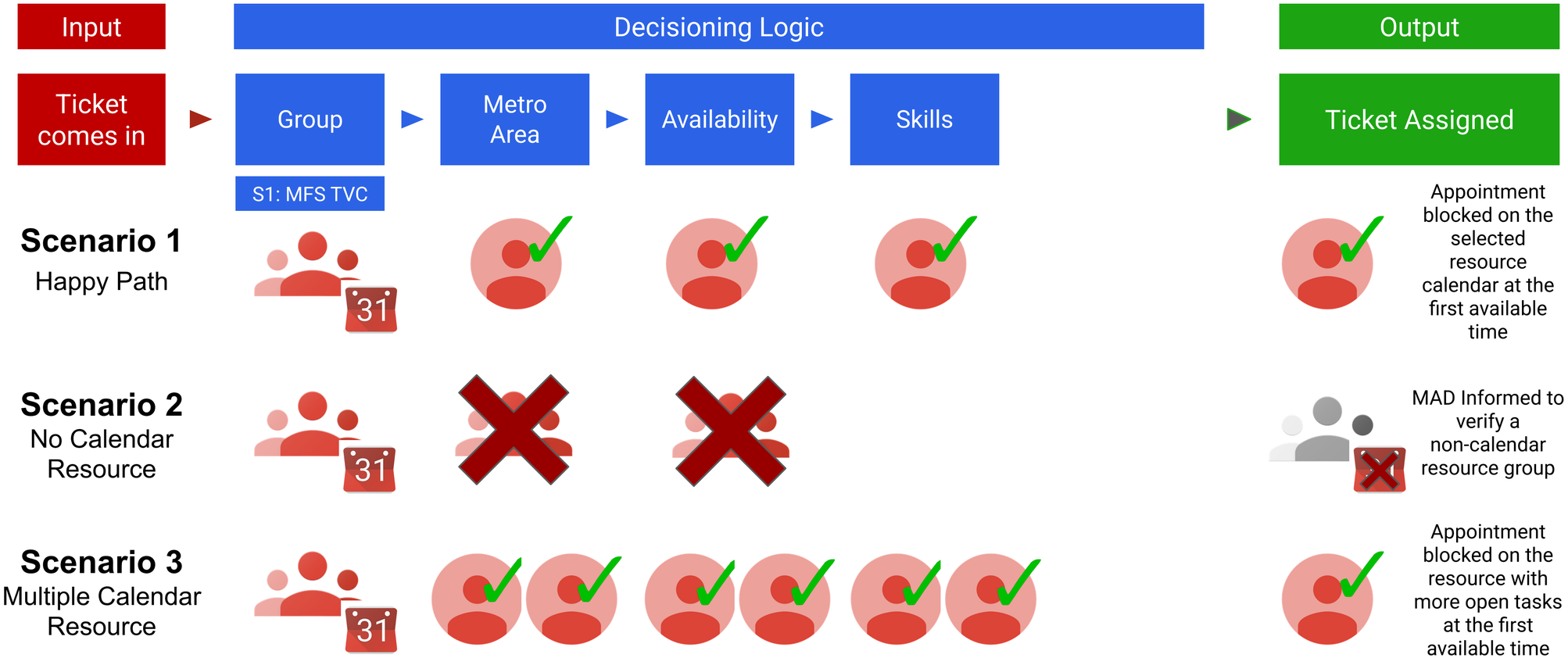


Pending Decisions - Engagement

	Outcomes
What do we do for QA for sites that require approval for photos?	Flag each question as RequiresPhotos? If a site does not allow photos, then hide that question
Which group is ultimately responsible for confirming / resolving tasks? Customers or Metro Owners?	Work Complete -> Confirmed AND PendingQA need to be complete ticket can be resolved. Proposal - WorkComplete -> QAComplete -> Confirmation
Are building mobile app first or desktop app first?	Planned for SPRINT 3
Are ALL resources having a task ENGAGEMENT view in Pega, with a user-specific DASHBOARD?	Planned for FUTURE SPRINT

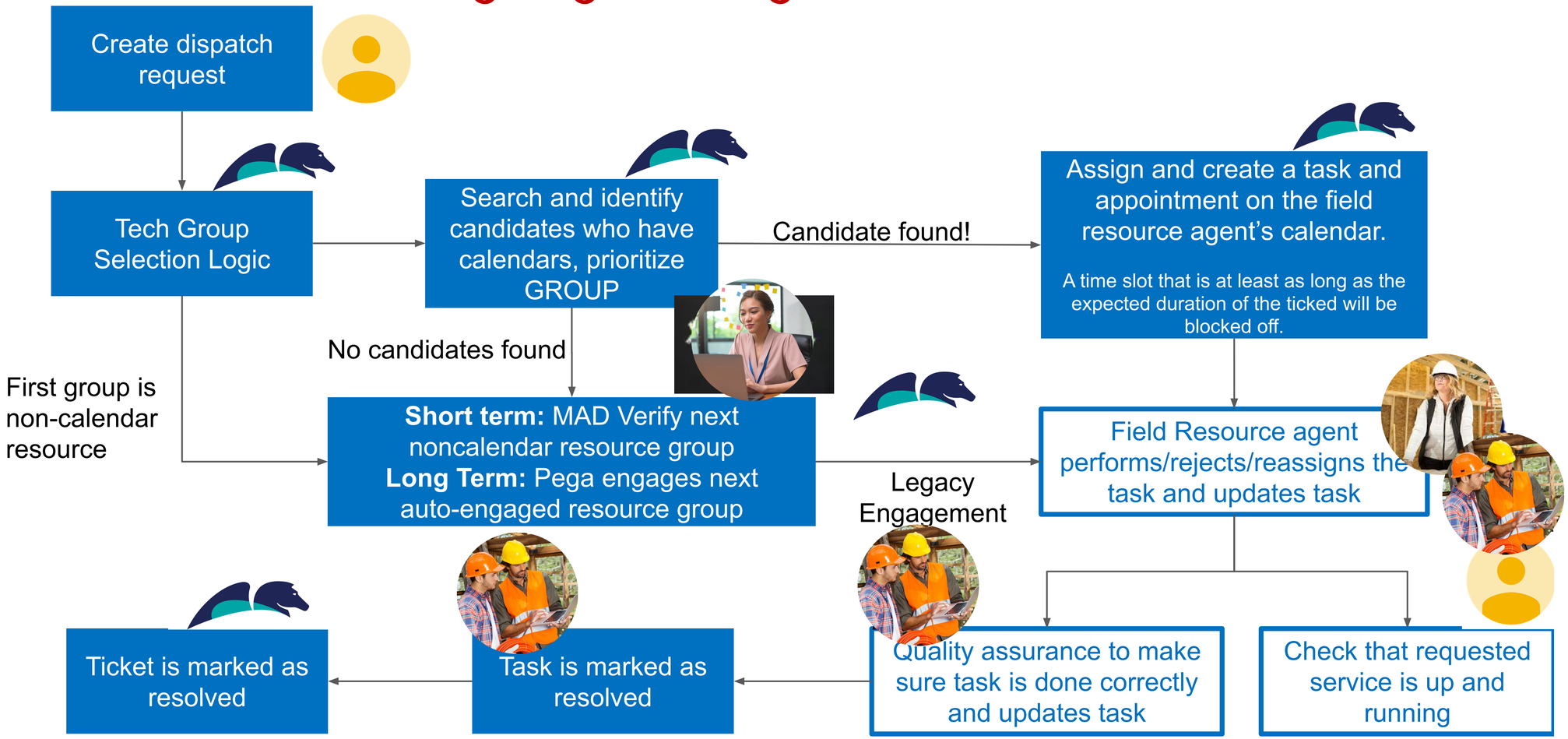
‘DESIGN THE PROCESS’ END DELIVERABLE

Sprint 1 Decisioning Scenarios



‘DESIGN THE PROCESS’ END DELIVERABLE

Decisioning Logic: End game - what we heard...



FIELD WORKER INTERACTION DESIGN

FIELD WORKER INTERACTION DESIGN

- Getting to know the user
- **Shadowing** them around for a day.
- Got to see an actual POP **site they visit**
- **Card sort exercise** for initial design of the ticket to understand the IA and prioritization of all data fields

SUGGESTIONS TO STAKEHOLDERS

Most of Moose was a **desktop enterprise**

For fieldworkers I suggested **MOBILE FOCUS**

Why?

Key Observations

Limited WiFi

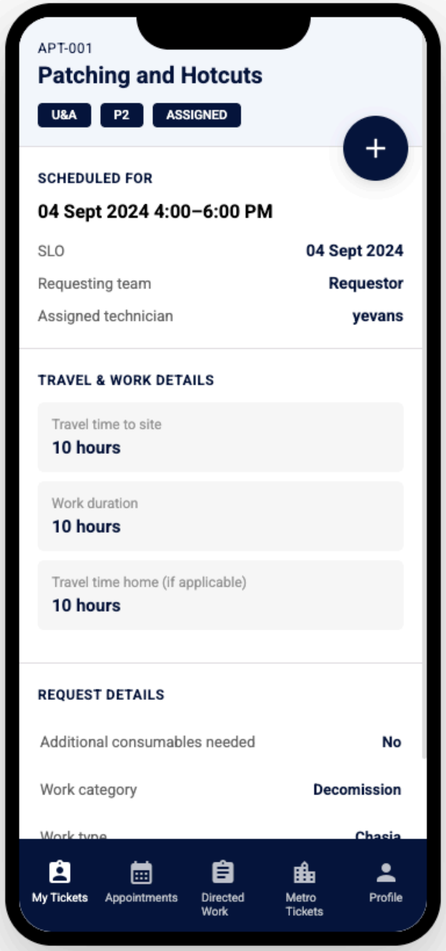
On ladders a lot! On the move

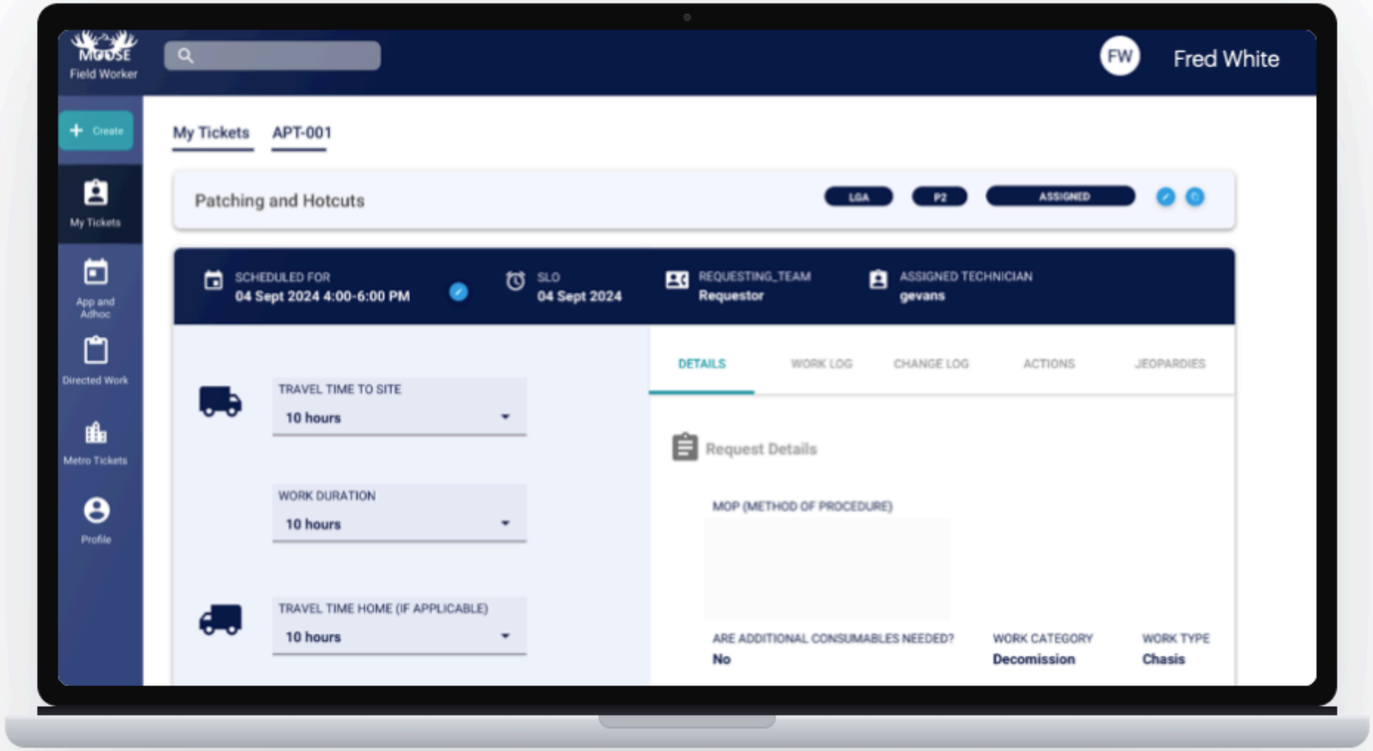
If they have a laptop, leave it in their truck in a backpack

Field Worker

Metro Owner

MAD Agent





ADAPTING DESIGNS BASED ON FEEDBACK

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Anonymous survey
every few weeks (Google Forms)

Field worker interviews
over Google Hangouts (across various metros)

Keep in mind **End User Dynamics** through out feedback process

ADAPTING DESIGNS BASED ON FEEDBACK

Field Worker

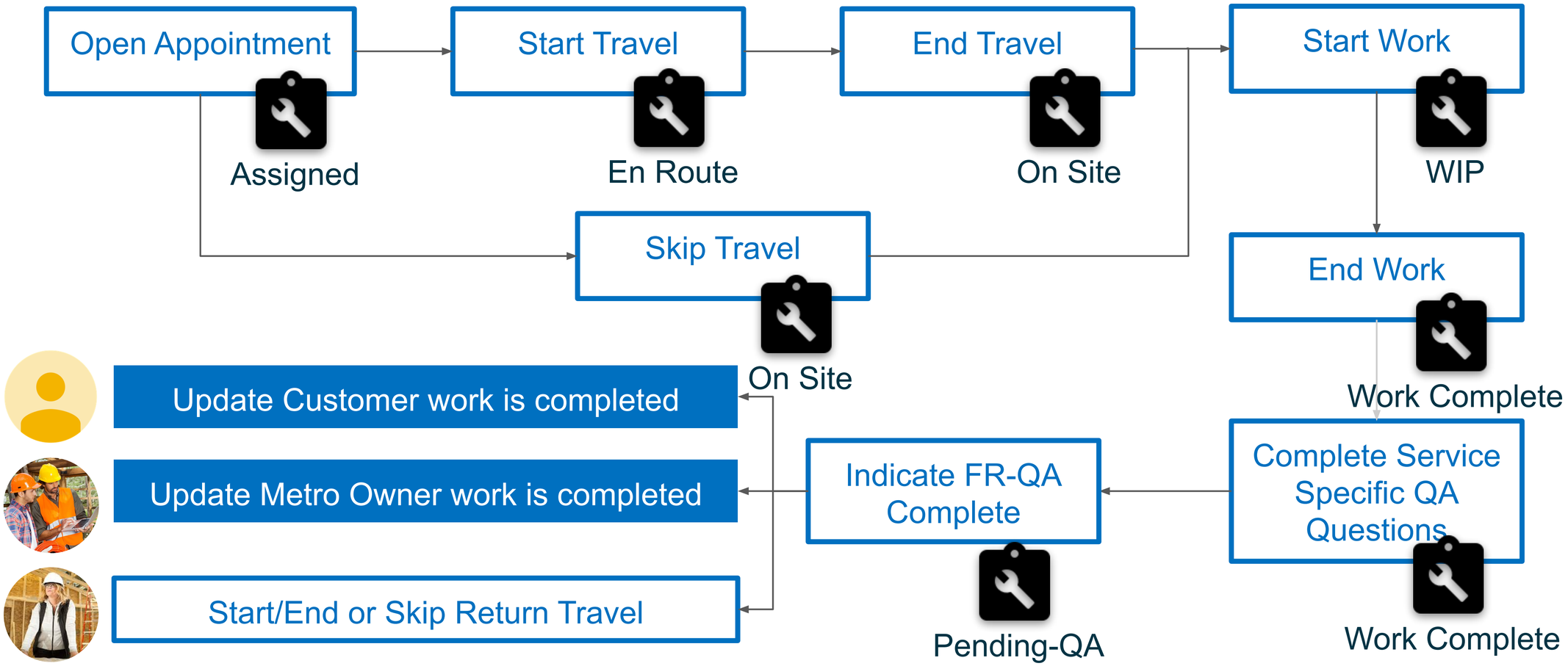
Metro Owner

Keeping in mind end user dynamics: **talked to field workers without metro owners present**? Why?

Power Dynamics: having your boss present effects the feedback

Metro owners are the **main client stakeholder but NOT the only user**

Today: Field Resource Performs the Appointment



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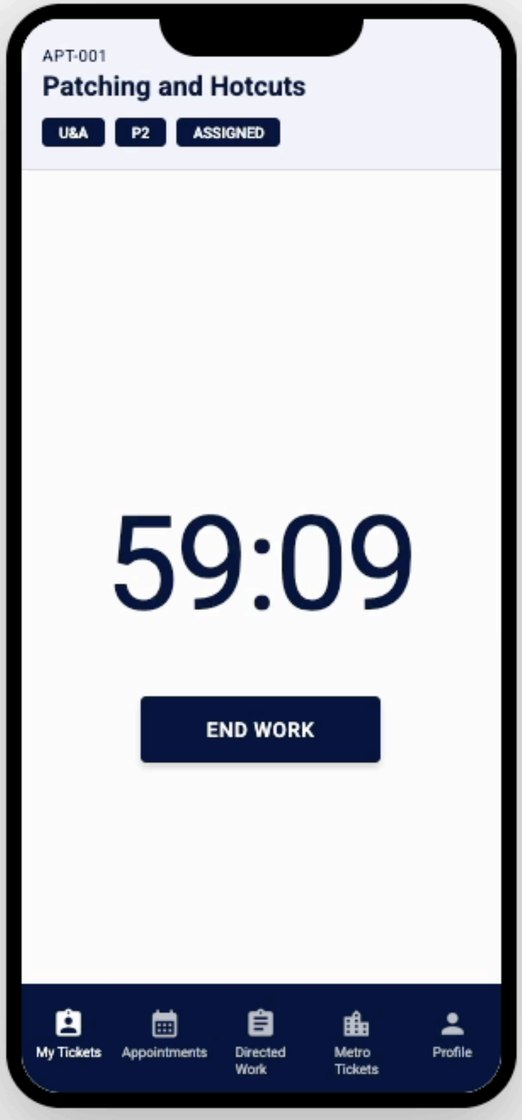
INITIAL DESIGNS



Field Worker Feedback

We'd like a 'pause work' button

Conducts 5 Whys exercise

Feels like we are being micromanaged

Having a timer ticking recording you to the second feels stressful

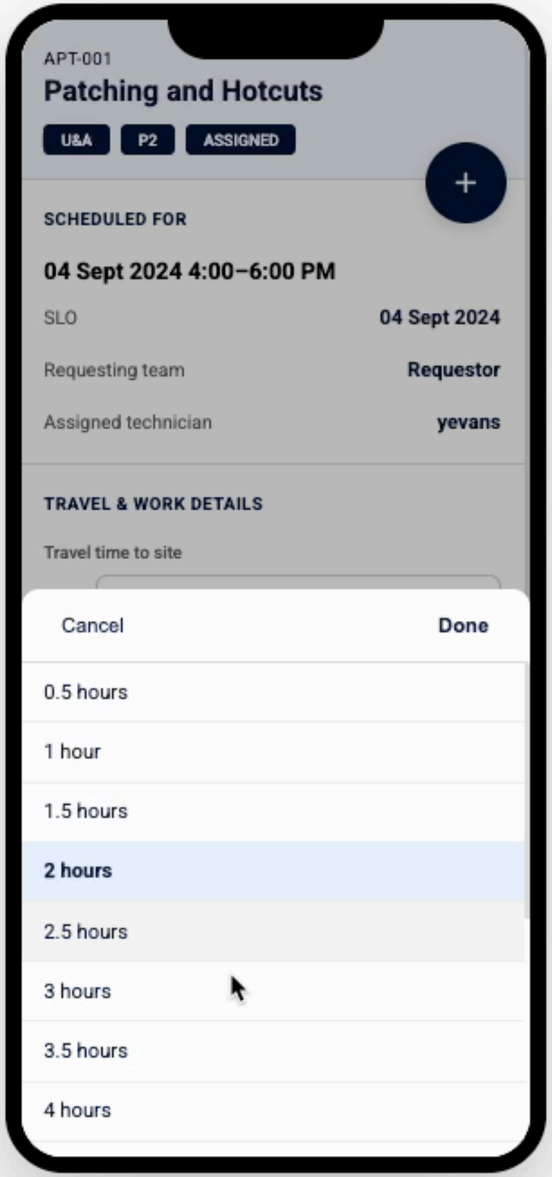
We don't always work on one ticket at a time. Sometimes, since I have the tools out already I'll do part of another job and then come back to the other one

ADAPTING DESIGNS BASED ON FEEDBACK

"Less anxiety inducing"

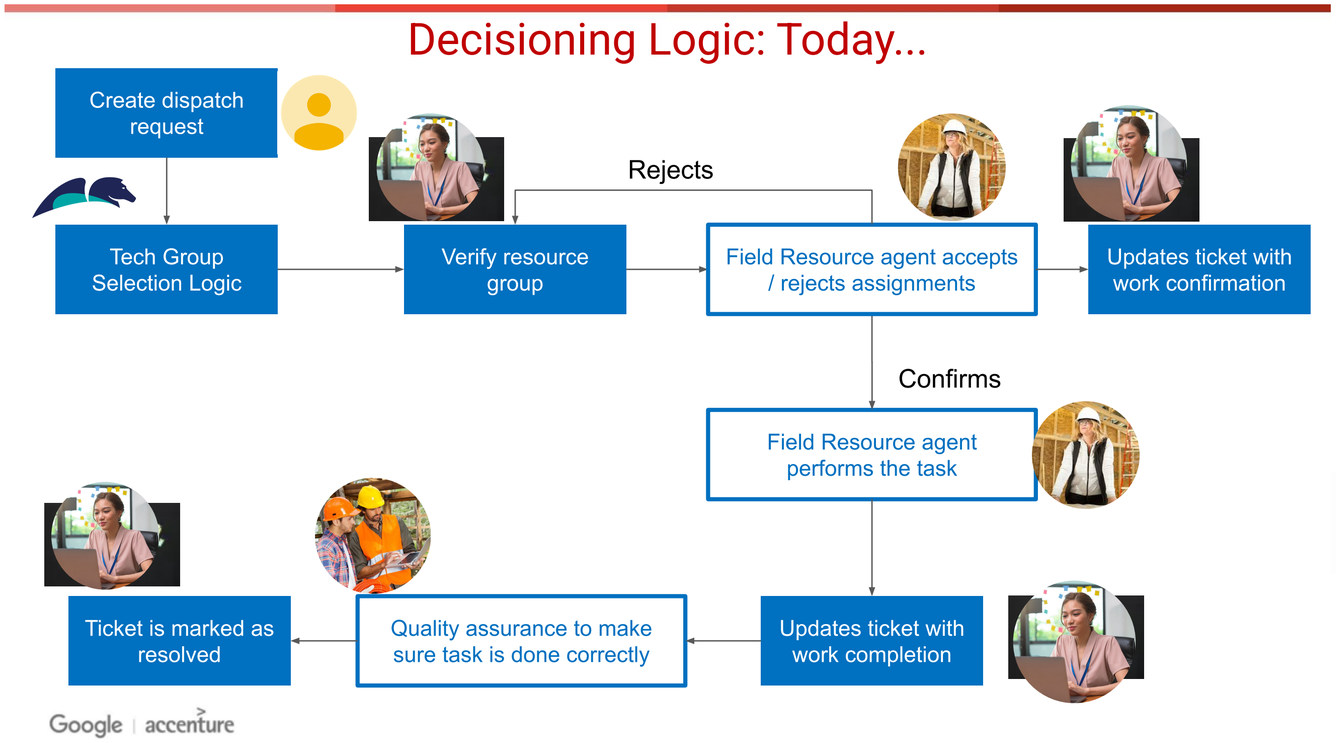
"More freedom"

"Feel like I'm trusted more"



"Less accurate but that's ok"

ADAPTING DESIGNS BASED ON PRODUCT VISION



Phase 1: Automation

Auto-assign field workers

Auto schedule appointments (rule-based)

MAD agent reviews and corrects

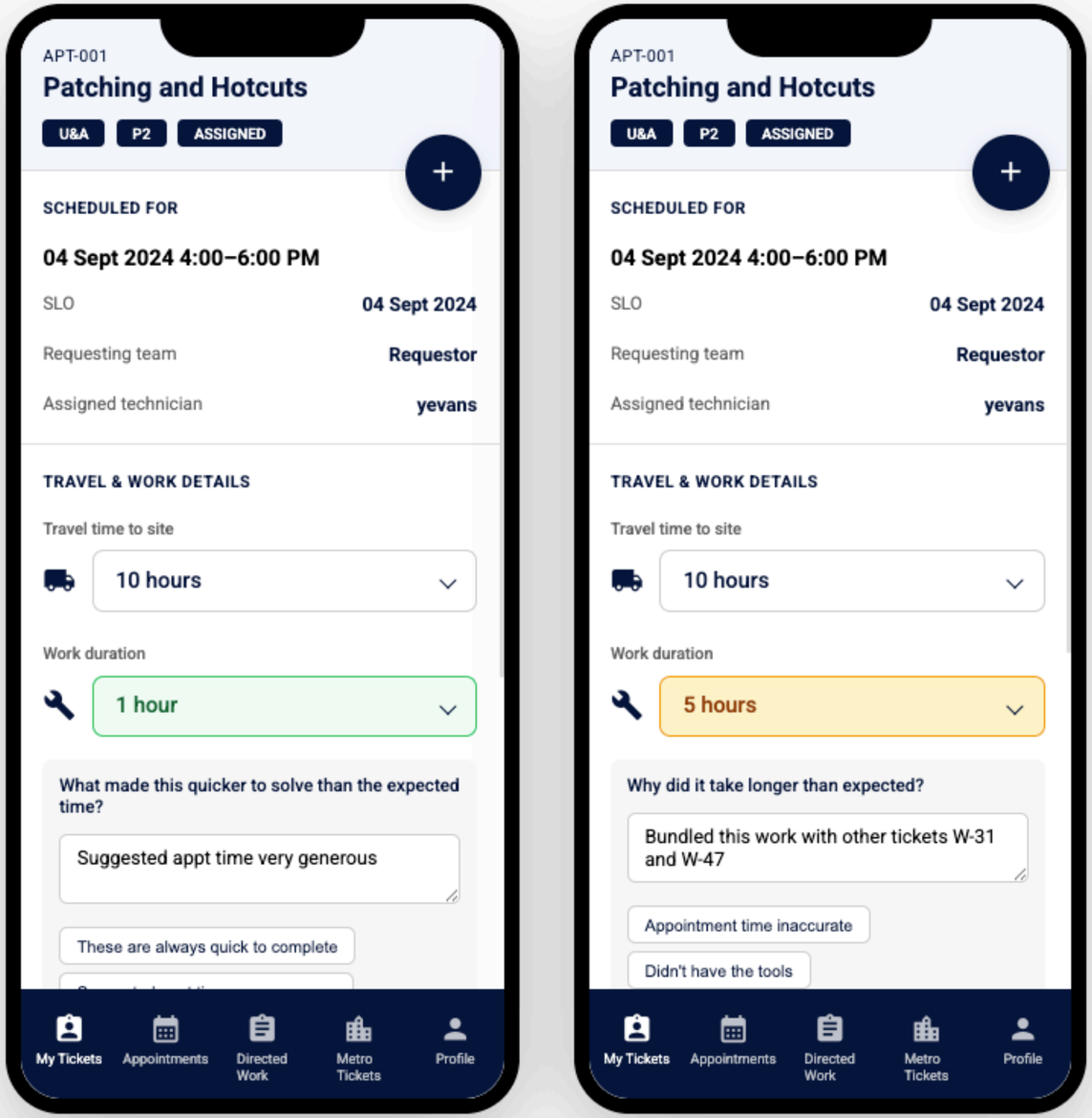
Phase 2: Data Informed automation

Assigns field workers by skill match

Sets appointment times from historical data

Minimal human oversight needed

ADAPTING DESIGNS BASED ON PRODUCT VISION



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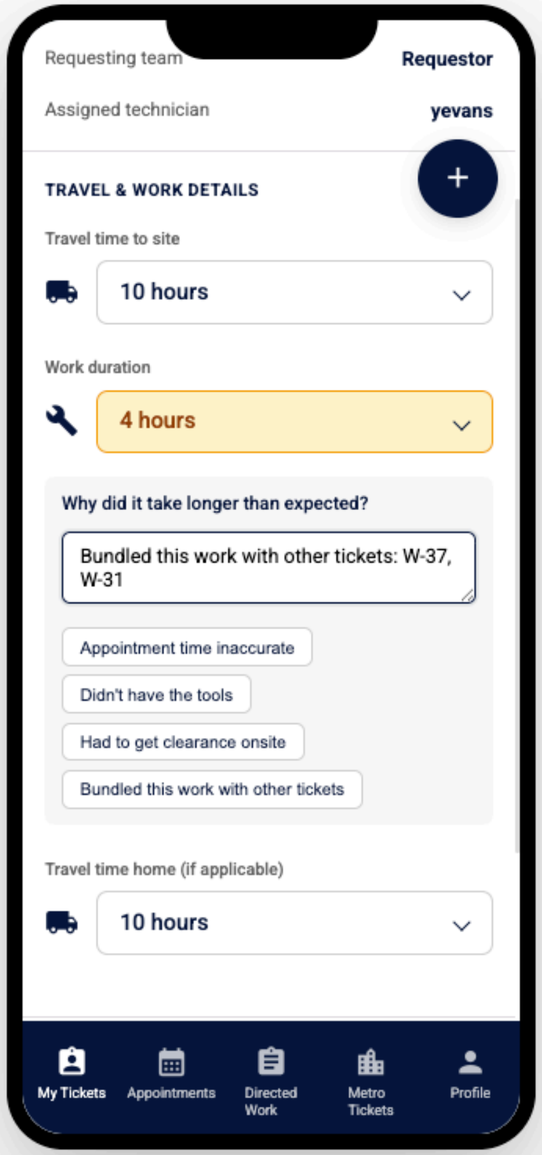

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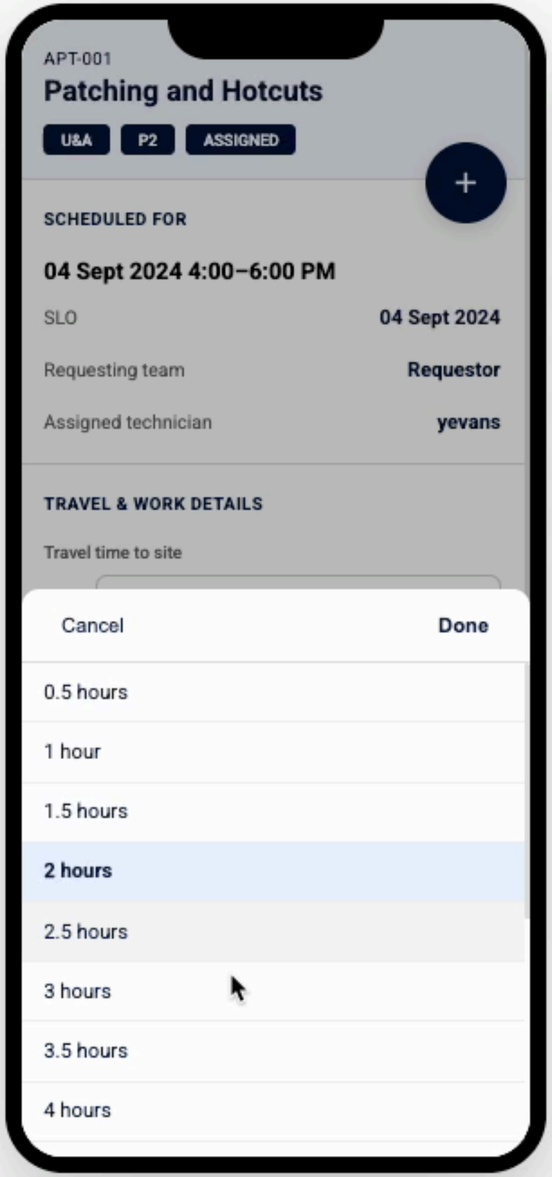


"Less accurate but that's ok"

"It would be a super nice feature to bundle "like" tickets together"

KEY METRICS OF SUCCESS

On a 1-5 scale of satisfaction, overall improvement by 3.5 points

Helped inform product roadmap

Led to contract renewal

“Instead of being watched I feel like I get to help make it better”

DESIGNING FOR EFFICIENCY

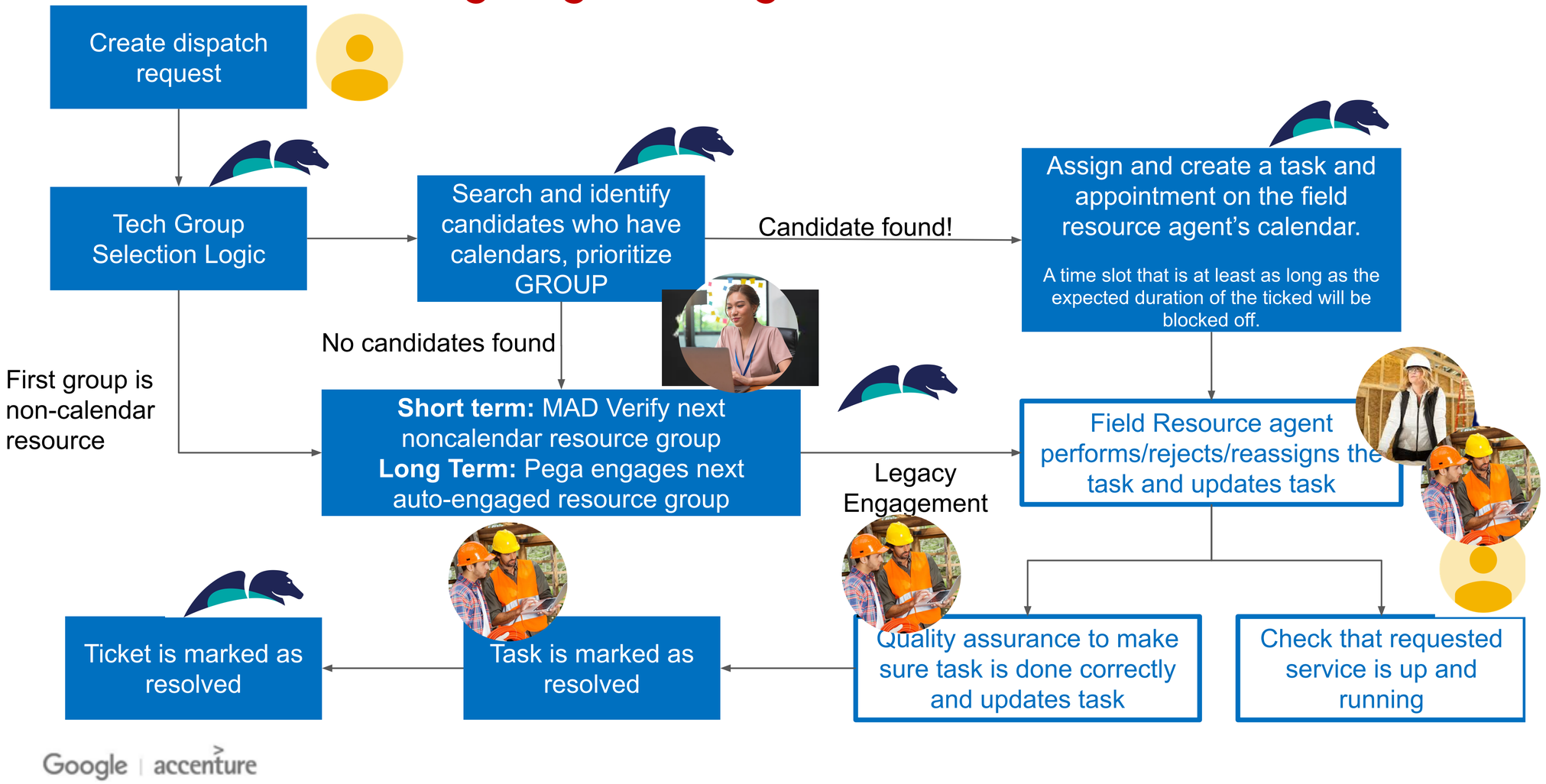
Field Worker

Metro Owner

MAD Agent

MAD AGENT INTERACTION DESIGN

Decisioning Logic: End game - what we heard...



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FROM INVISIBLE TO AT-A-GLANCE

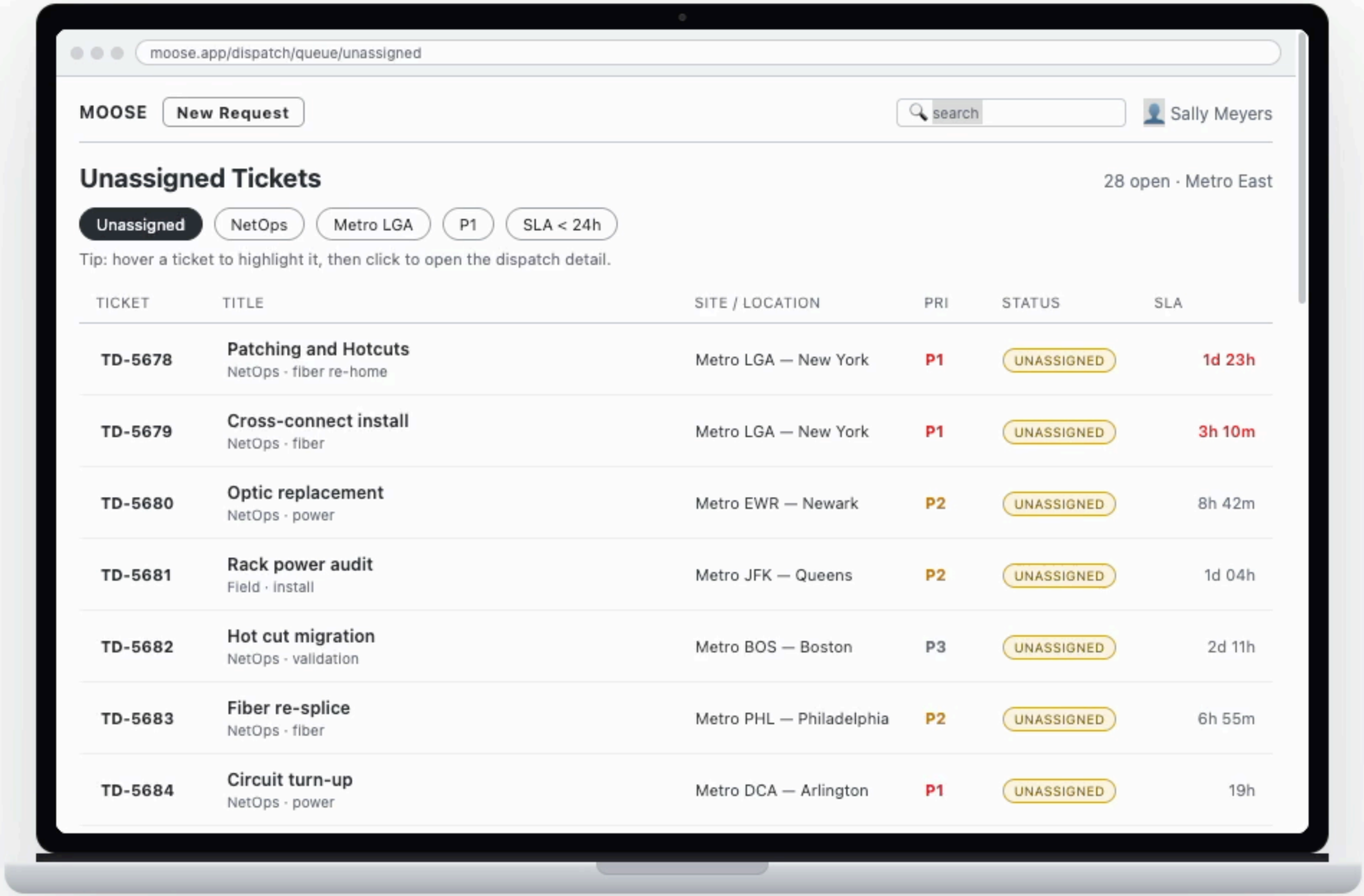
MAD Agent

Completely remote from the work site

Also in a different time zone

Must action the ticket in **under 6 minutes**

USABILITY WALKTHROUGH AND HEURISTIC EVALUATION



10 MAD Agents Interviewed

Manila

Frankfurt

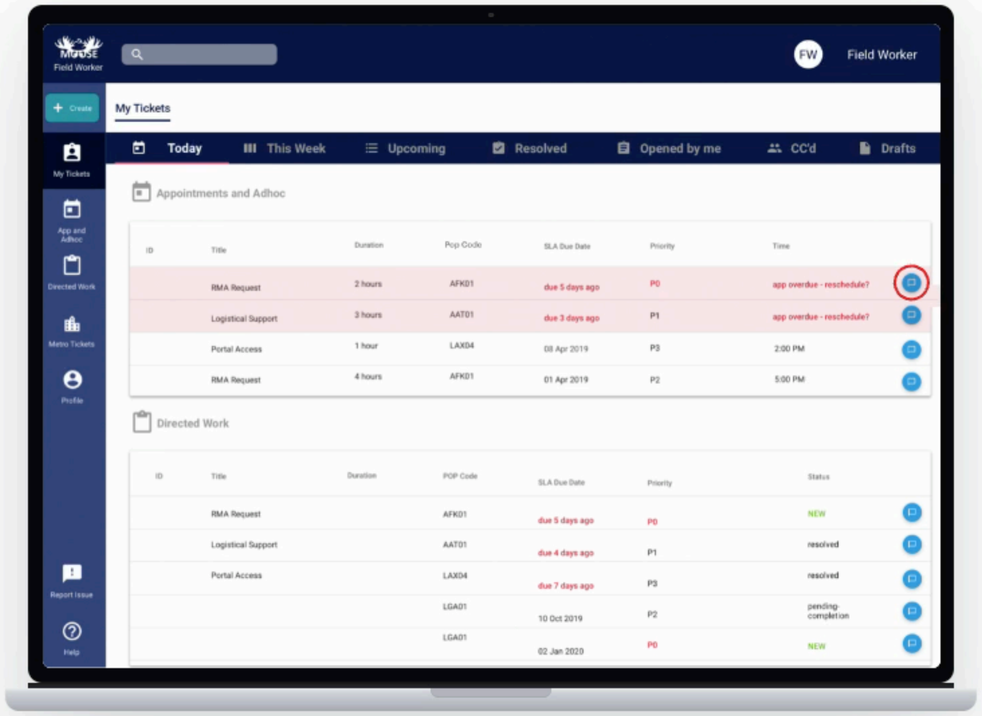
London

Key Insights:

Mostly use work log

Must click into the ticket and scroll all the way down to find it

END RESULT: SLIDE OUT WORK LOG



Pega doesn't have a slide out pane component.

Can you build a modal?

Yes.

Can you have it pinned to the right and take up the whole container height?

Yes.

KEY METRICS OF SUCCESS

Minimize number of clicks from 3 to 1

No more scrolling

On a 1-5 scale of satisfaction, overall improvement by 2.5 points

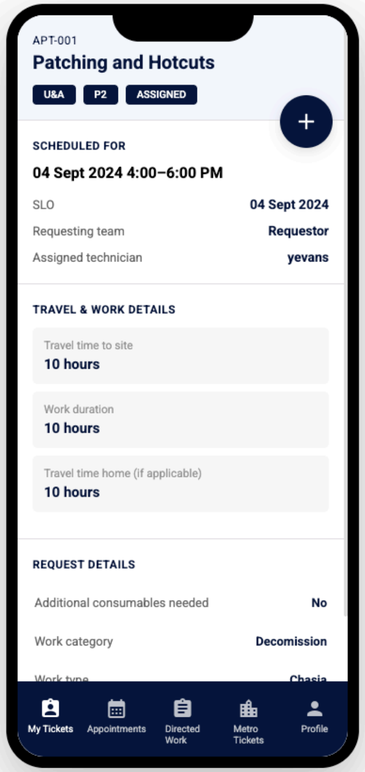
Ticket assign time down by 30%

“Slide out work log is very powerful and will save us a lot of time”

AGENDA

Google NetOps

Moose



Metro Owner platform + field worker mobile app that made automation possible across **10** workstreams, **4** timezones, and **3** user groups. **30%** productivity gain.

MICROSOFT EDUCATION

2020
Pandemic

Paradigm shift

WHY?

MS EDU was at the forefront of hybrid learning

MICROSOFT EDUCATION

Career Coach

Reading Coach

Learning Accelerators

Search Coach

Speaker Coach

Math Coach

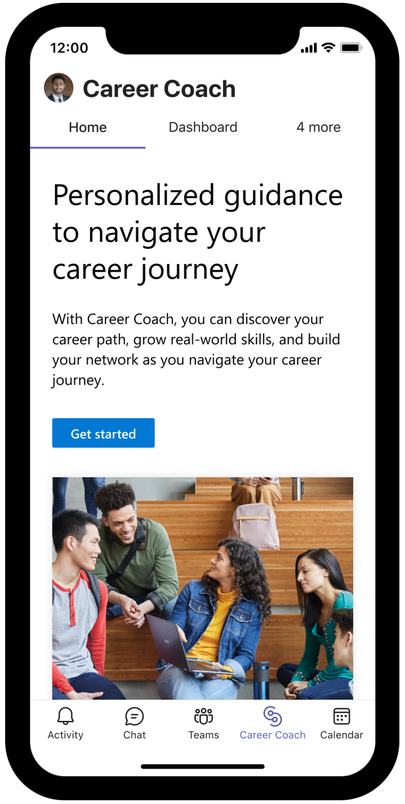
CAREER COACH

Showing Insights work on customer calls led to **three new universities signed** before GA launch. Huge market differentiator

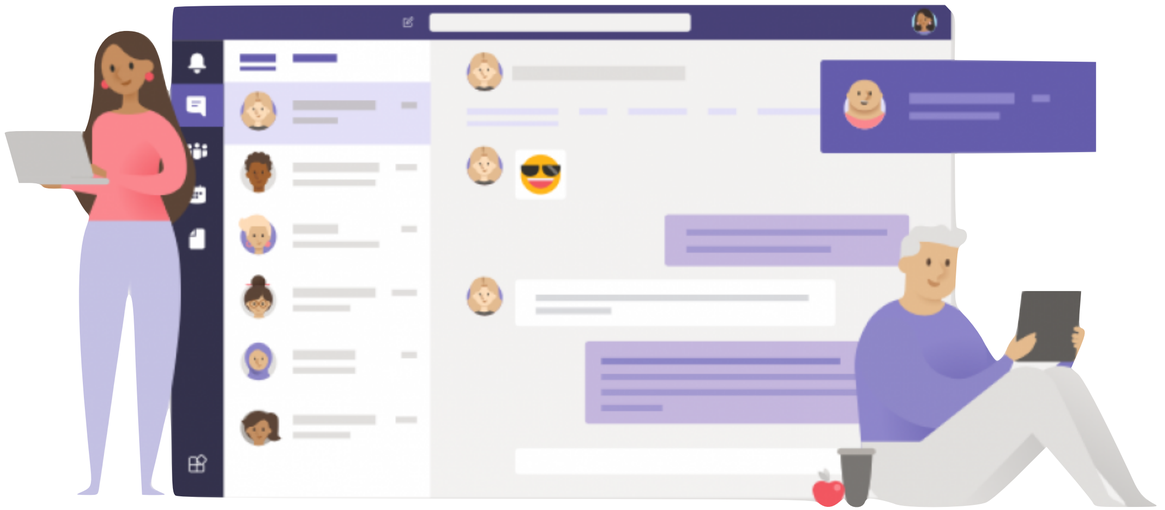
Increased usage of LinkedIn's JMI Economic graph by **32%**

LinkedIn agreed to work on combining ID work; shared long term goals

Increase active sessions for faculty (up **20%**)



MICROSOFT EDUCATION



Backbone EDU APPS

Assignments

Insights

Connections

MICROSOFT EDUCATION

Assignments

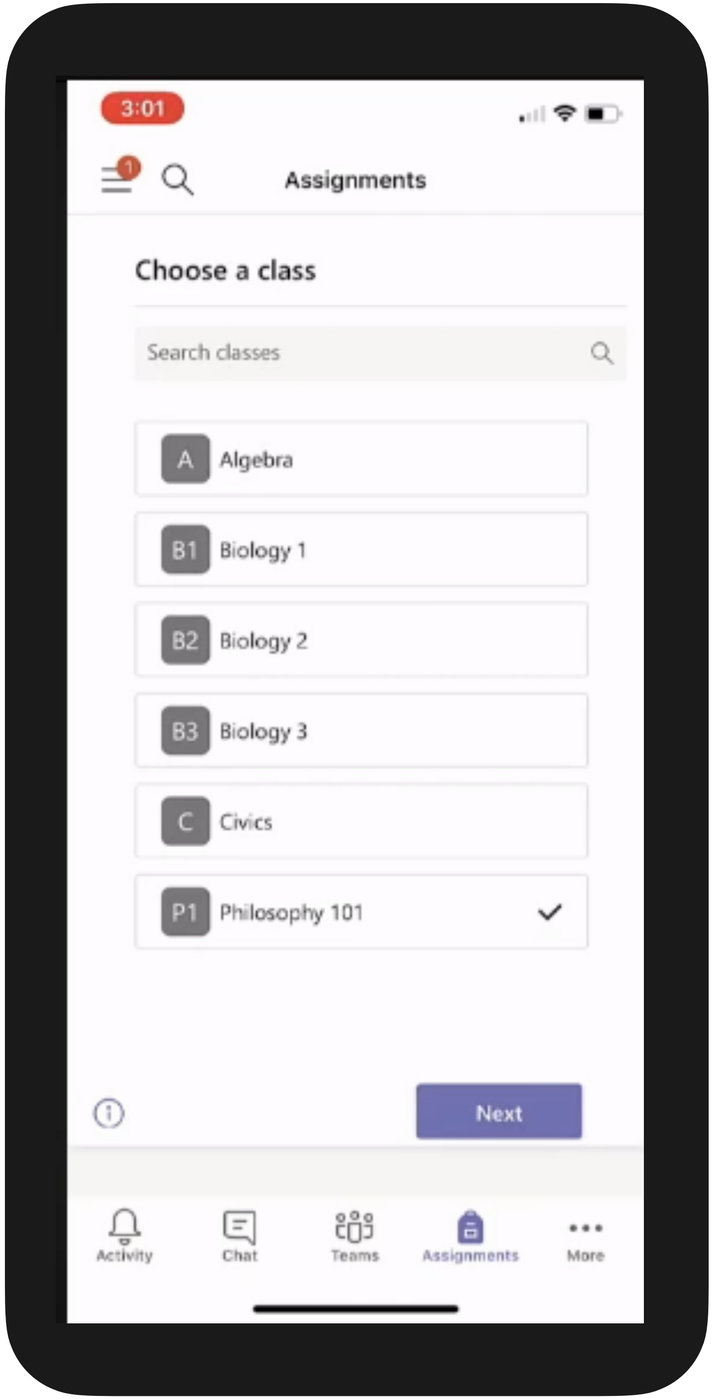
Due Date
Grade
Type ie.

Vanilla assignment

Quiz

Or a **LEARNING ACCELERATOR**

- **ie. Speaker Progress Assignment**
- **ie. Reading Progress Assignment**



MICROSOFT EDUCATION

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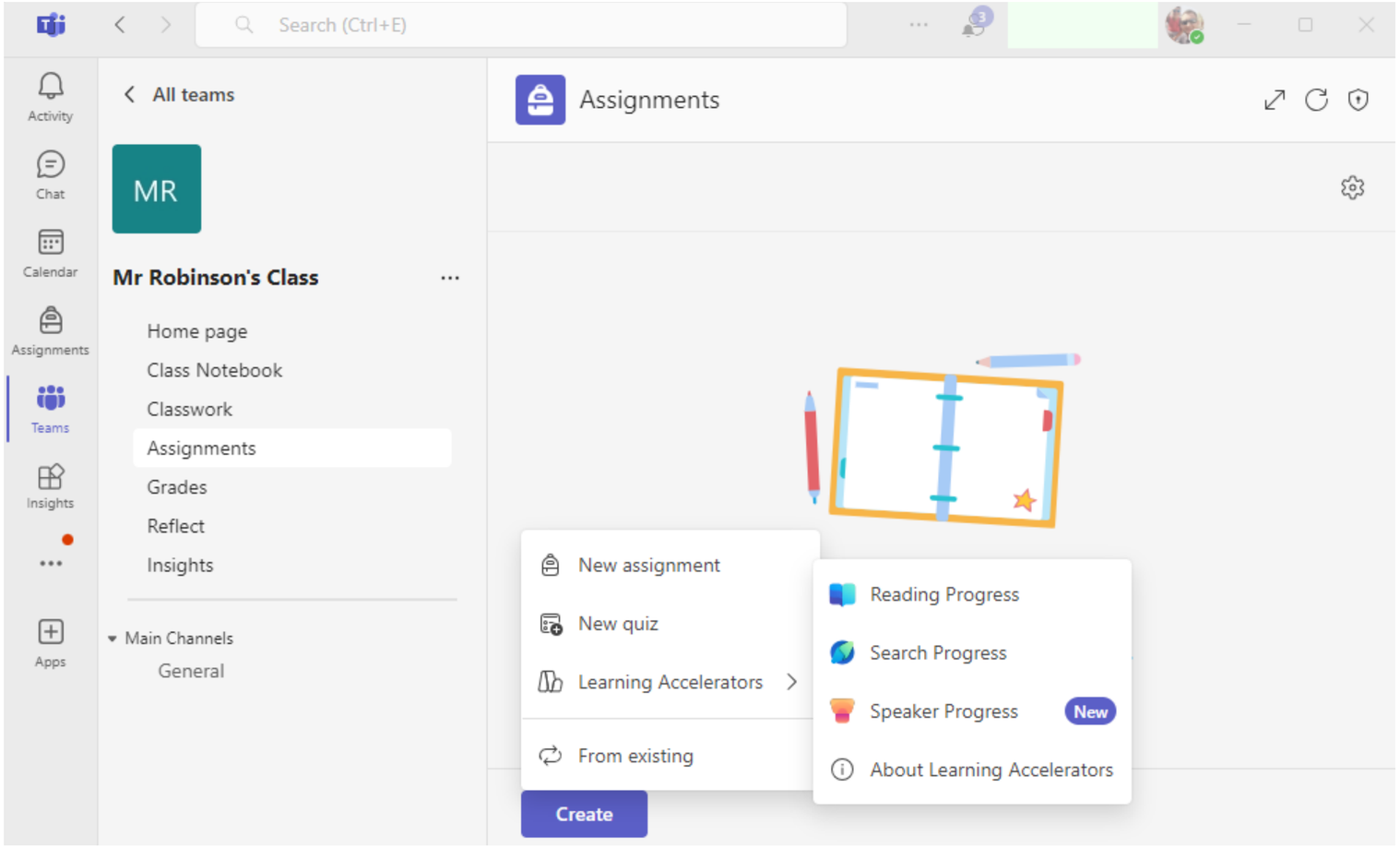


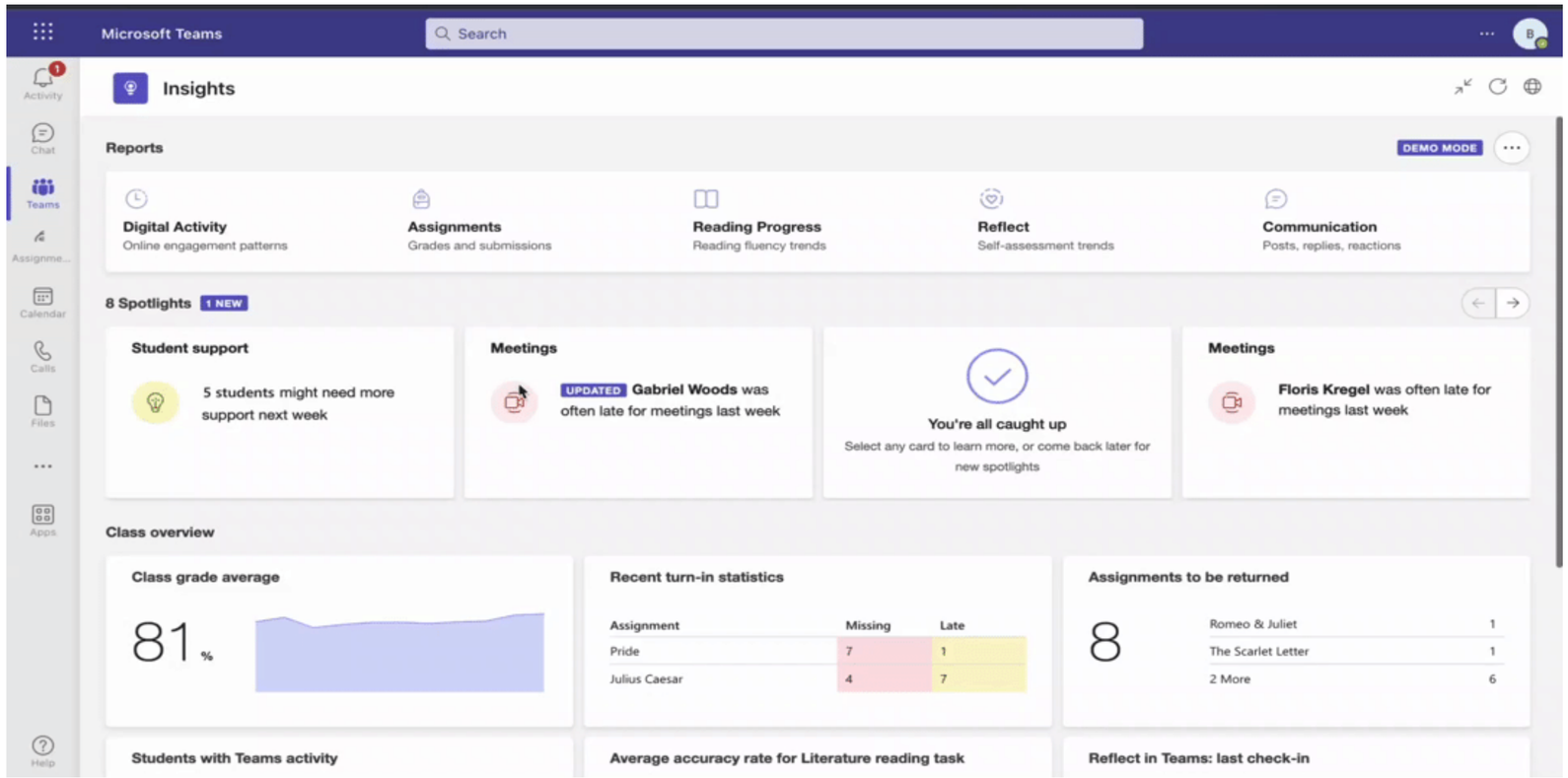
MICROSOFT EDUCATION

Insights

Historical data from past assignments

Pulls data from Assignments (the product)

Data from Insights (the Product) is also surfaced inside Assignements



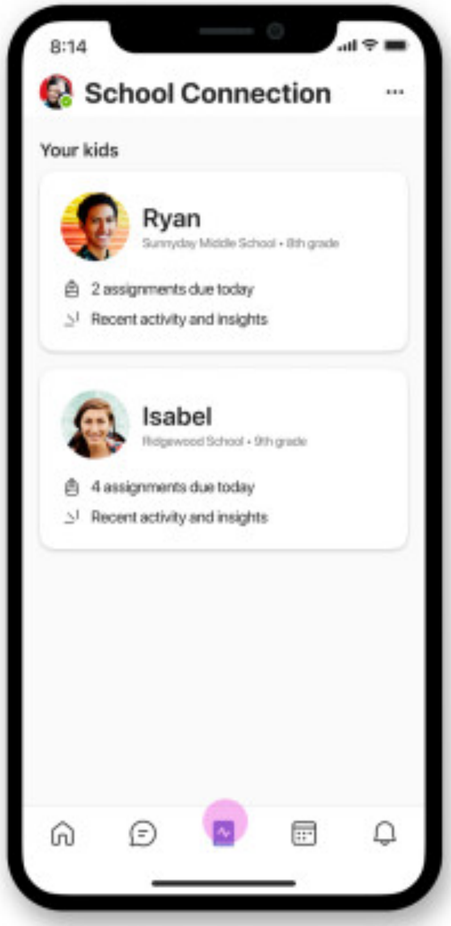
MICROSOFT EDUCATION

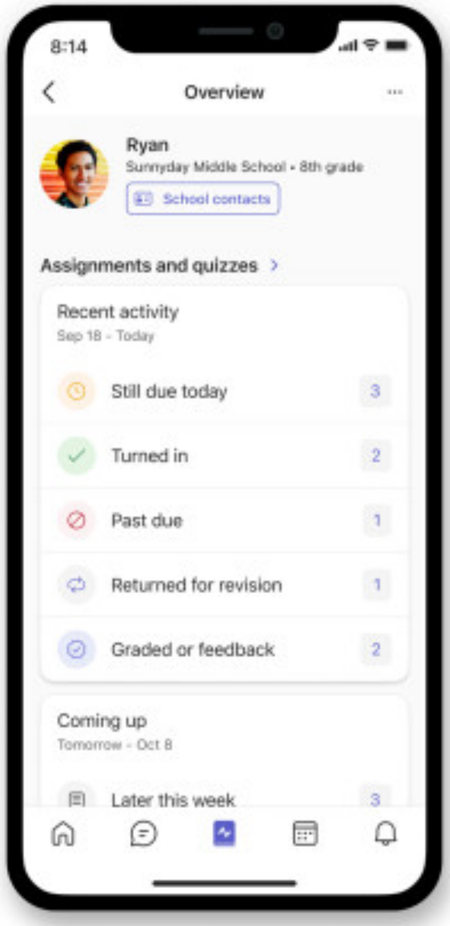
Connections

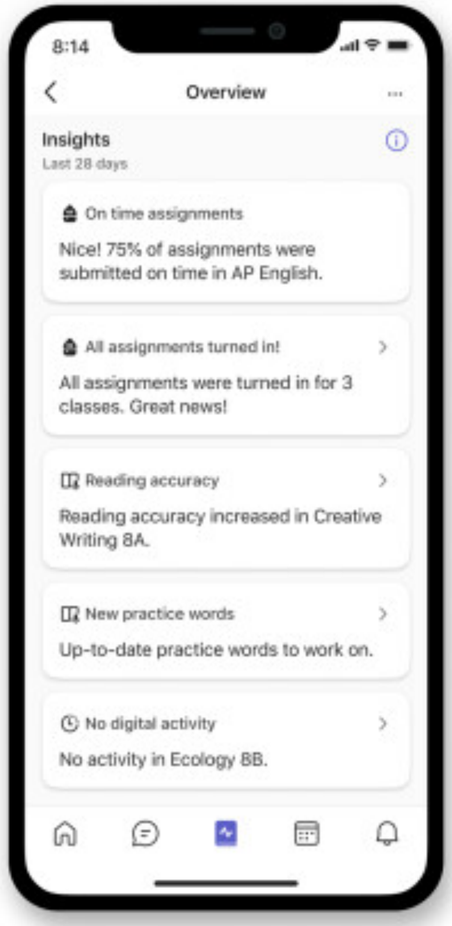
Parent app

Lets them see how students are doing

- emotionally
- academically

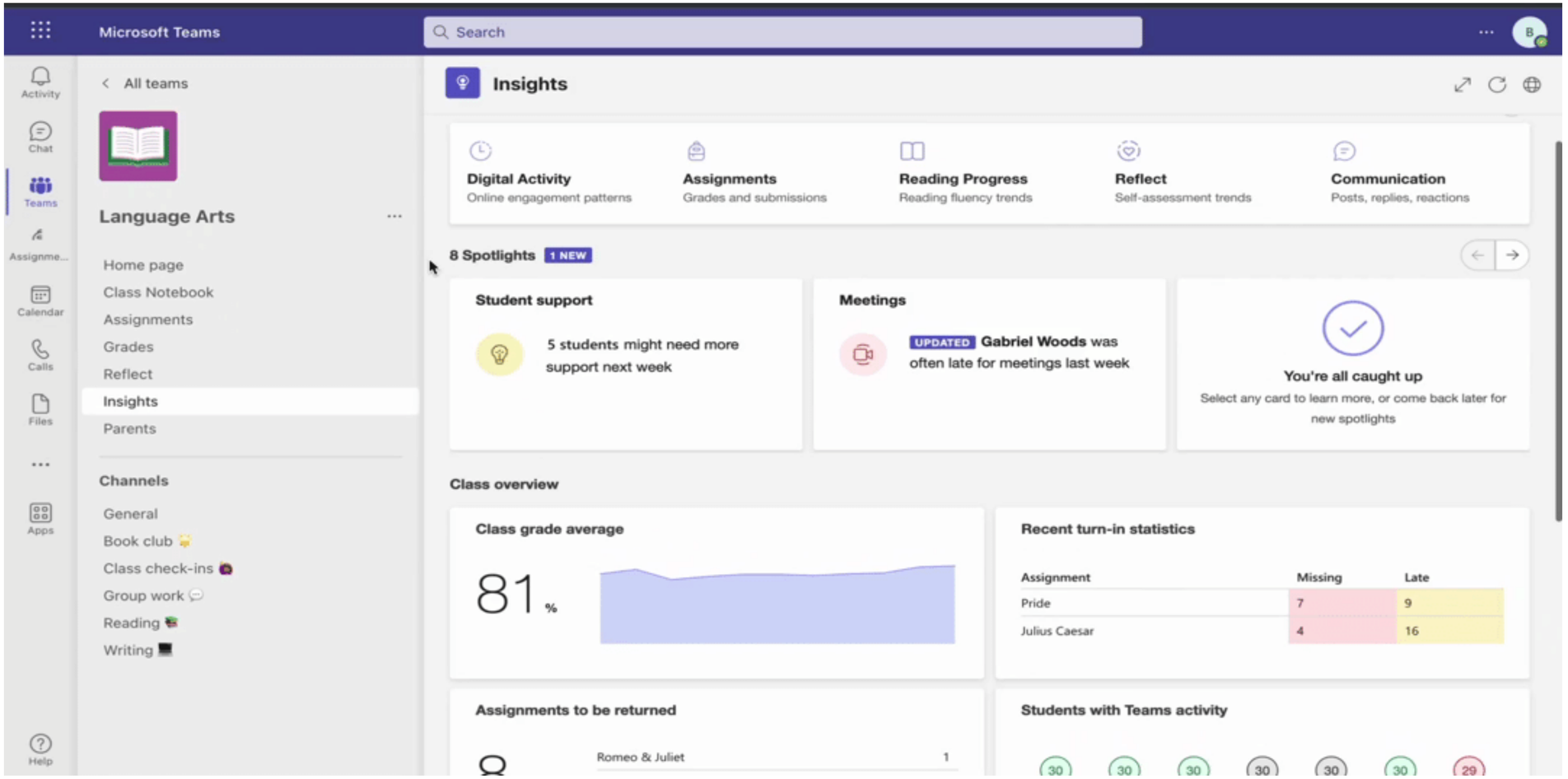






MICROSOFT EDUCATION

How might drive traffic to Insights (the app) from Reading Progress (or any Assignment type)

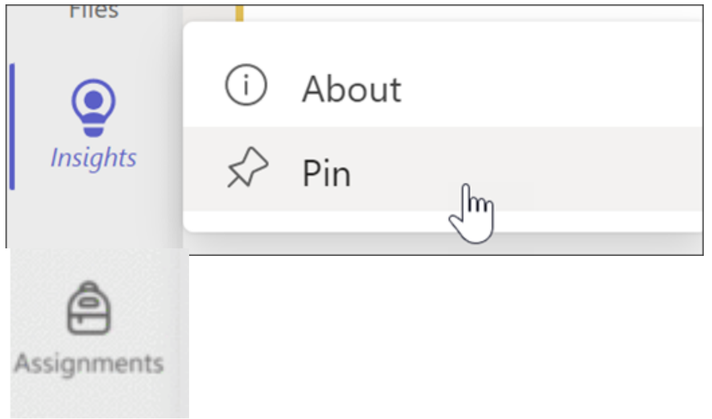


INSIGHTS DISCOVERABILITY FROM ASSIGNMENTS

How might we drive traffic to Insights (the app) from Reading Progress (or any Accelerator)

Key Partners

- Insights designers + product team



Very motivated partner, why?

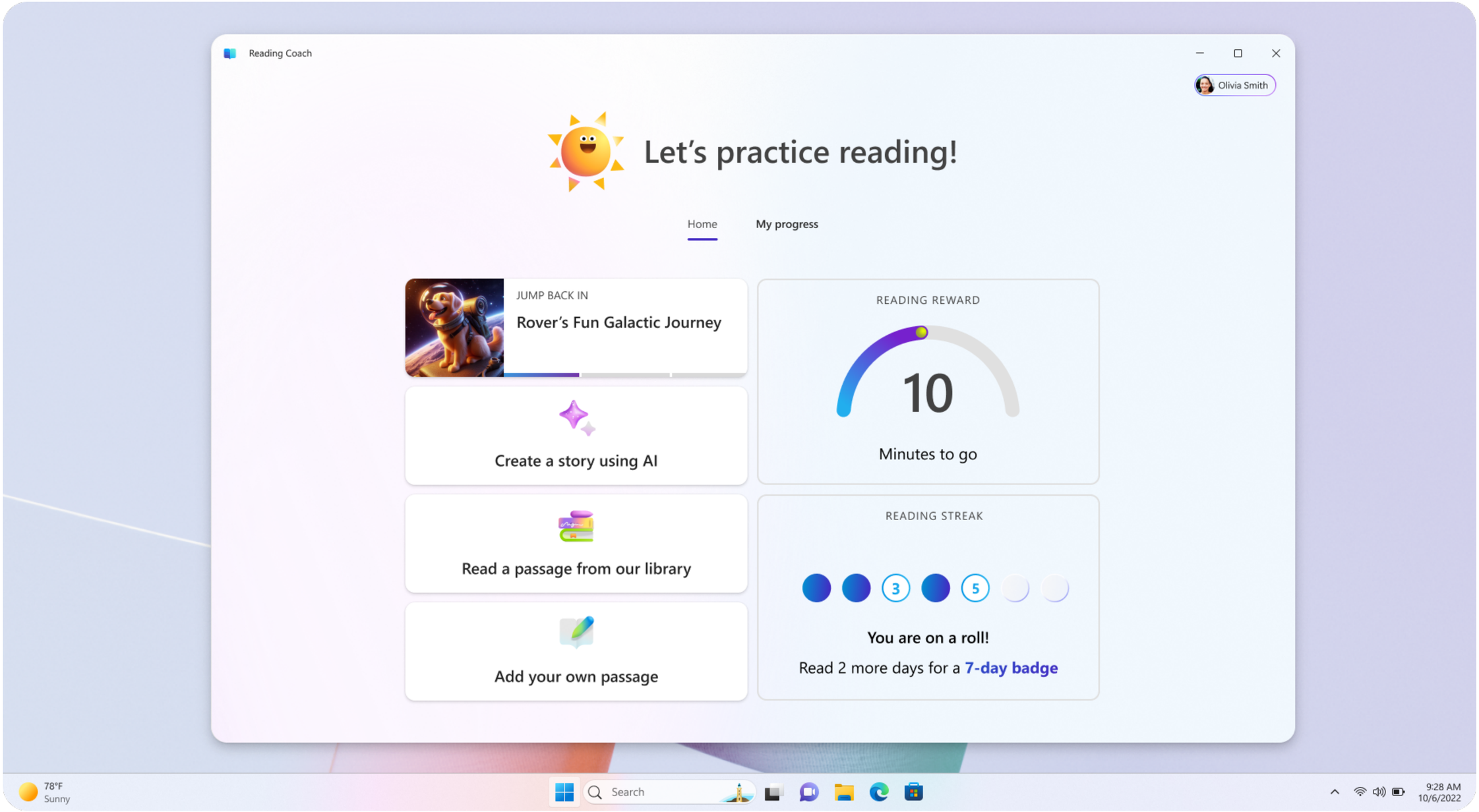
- Want to drive more traffic to their product, Insights

INSIGHTS DISCOVERABILITY FROM ASSIGNMENTS

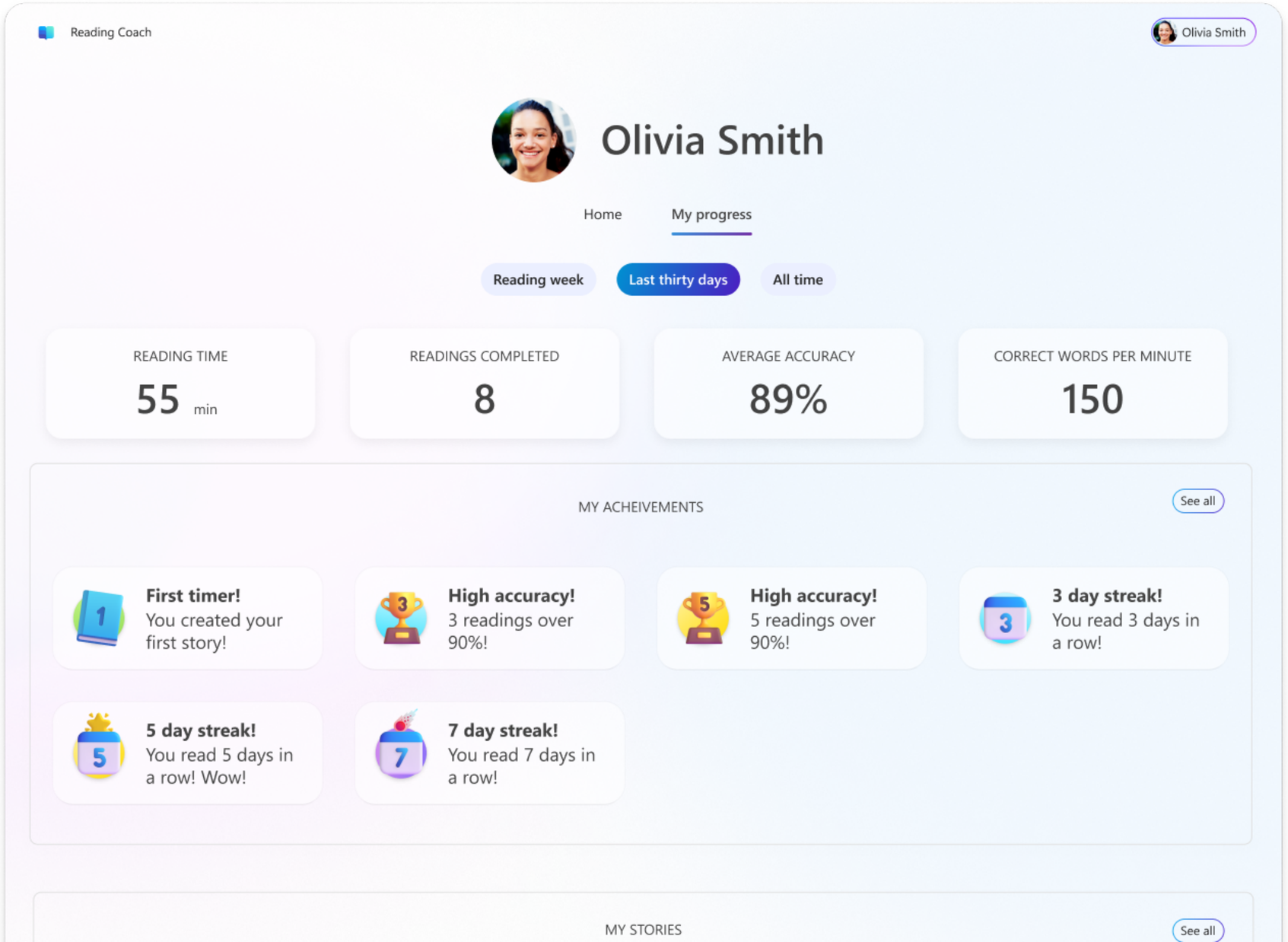
Discovery

- Establish where we are showing Insights relevant data across all Assignments ie. Reading
- Establish audience, students? teacher? Both?

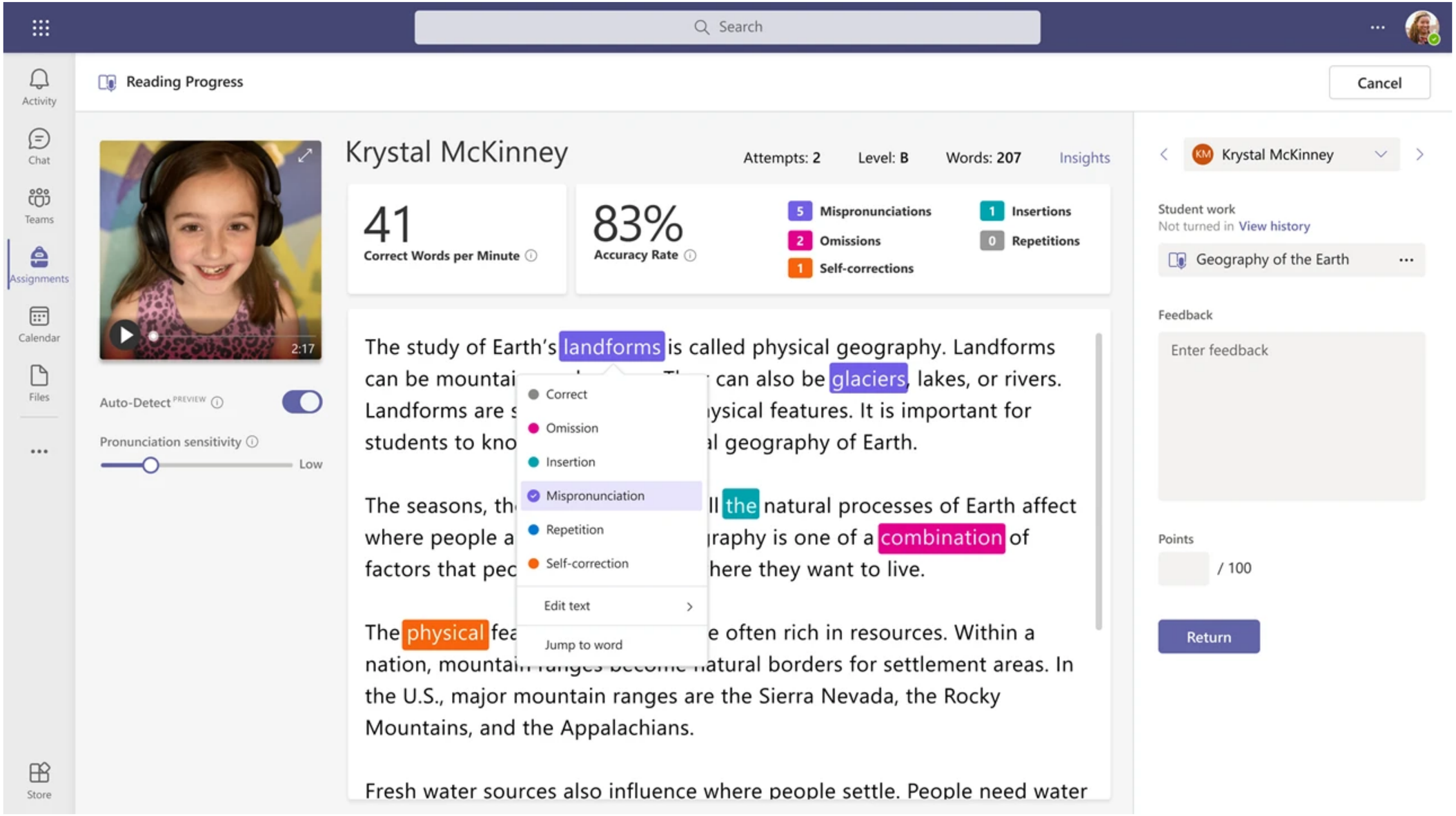
INSIGHTS DISCOVERABILITY FROM ASSIGNMENTS



INSIGHTS DISCOVERABILITY FROM ASSIGNMENTS

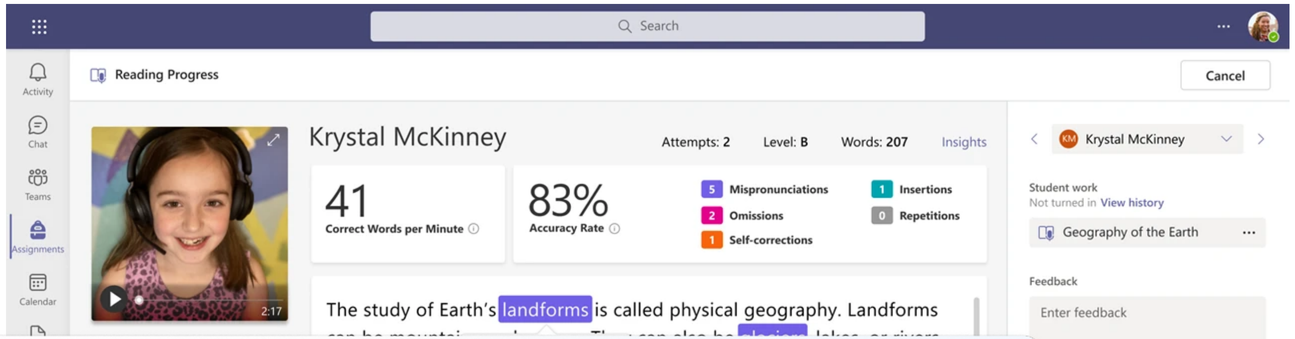


INSIGHTS DISCOVERABILITY FROM ASSIGNMENTS

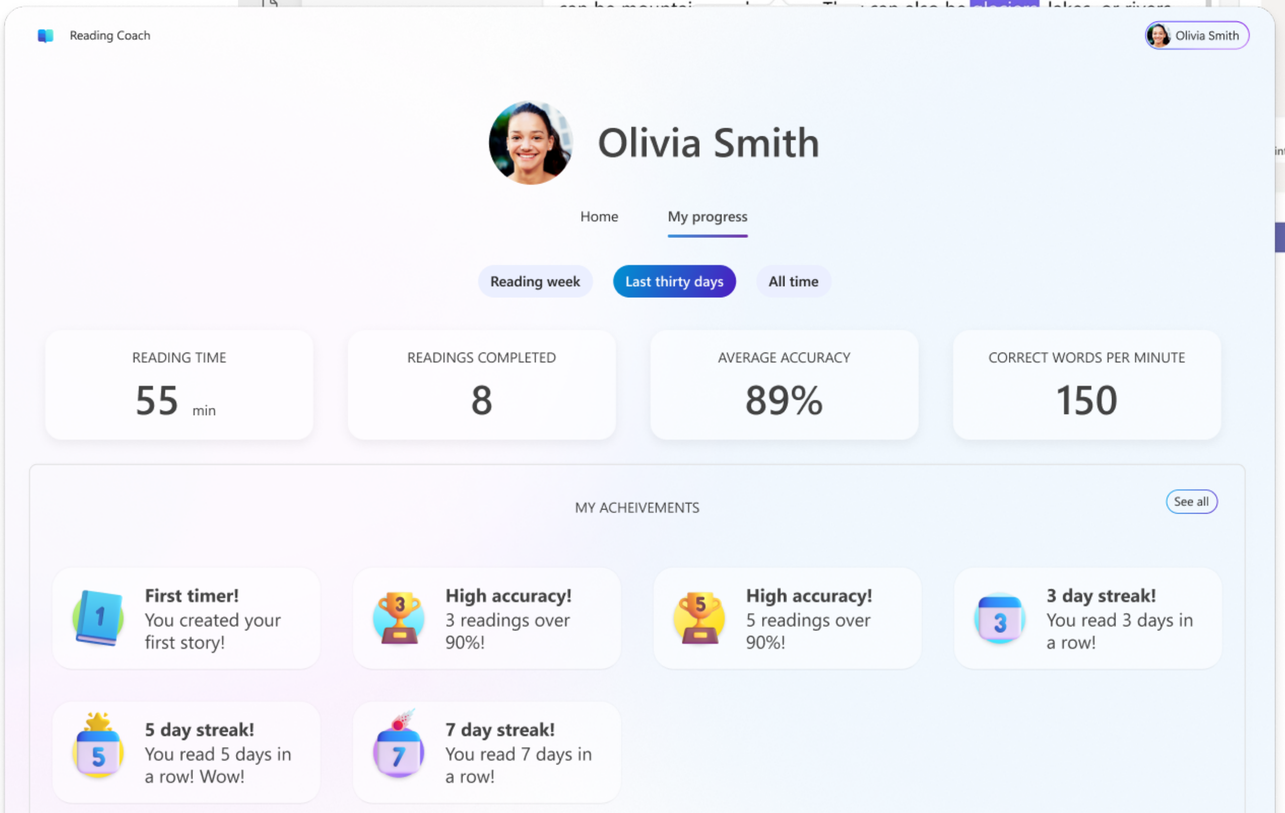


INSIGHTS DISCOVERABILITY FROM ACCELERATORS

Classroom Experience



Standalone



Establishing Container

Focus on Classroom experience (ie Reading Progress) instead of standalone (ie. Reading Coach)

Why? Insights team confirmed they don't have standalone users -> mostly schools

INSIGHTS DISCOVERABILITY FROM ACCELERATORS

Interviewing teachers

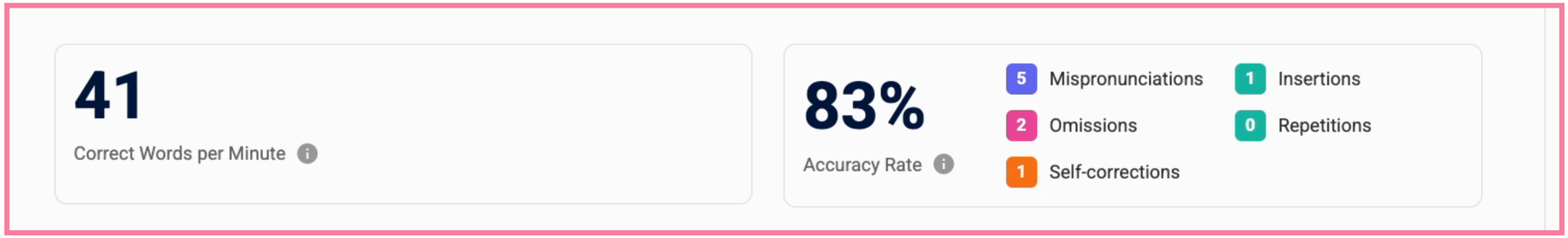
- What type of data do you care about for your students?
- Trends? Highest score yet? Comparing their historical data versus class average

Key Insights:

Care about **outlier data.** Ie. They usually score a 90% but this time they scored lower 20% lower than their average

The grade itself isn't important, **context of the grade** is more important

INSIGHTS DISCOVERABILITY FROM ACCELERATORS

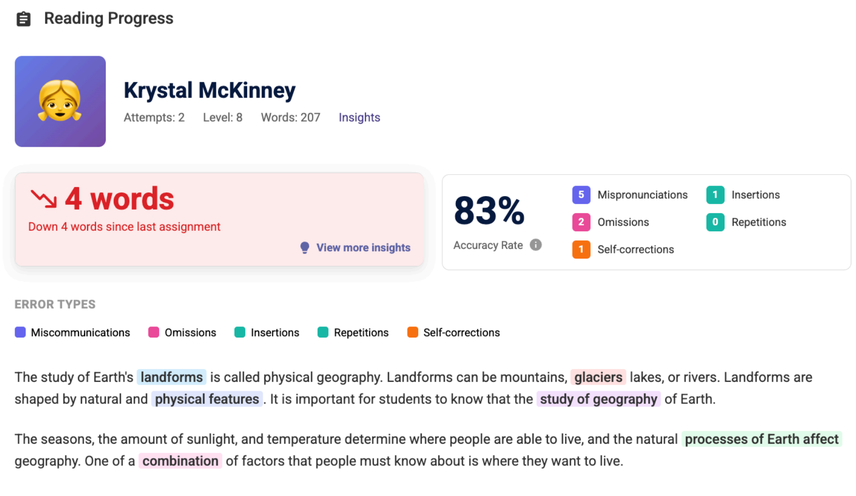


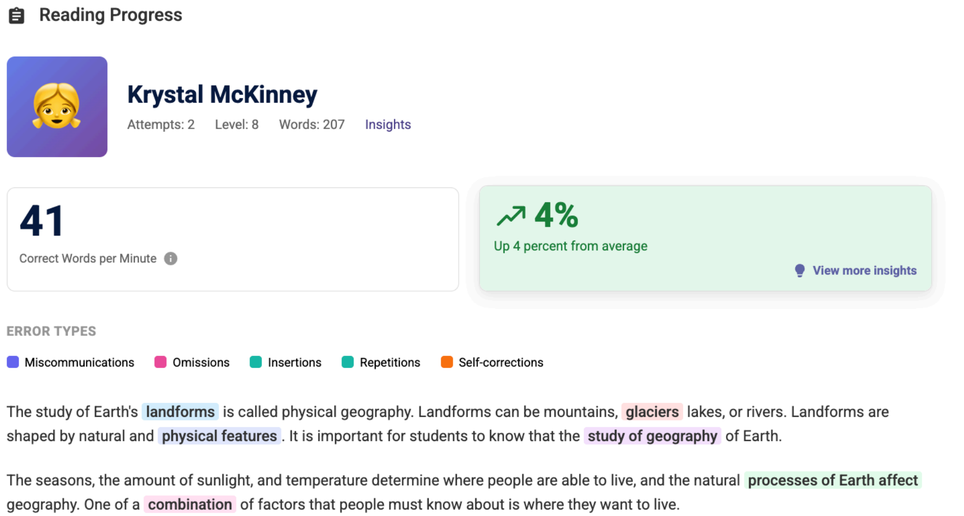
Interaction Design Ideas

Info icon already next to labels - has a definition of the data label meaning- what if we add insights there?

What if we flip the whole card?

INSIGHTS DISCOVERABILITY FROM ACCELERATORS



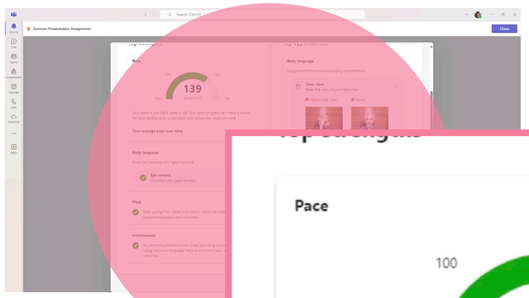


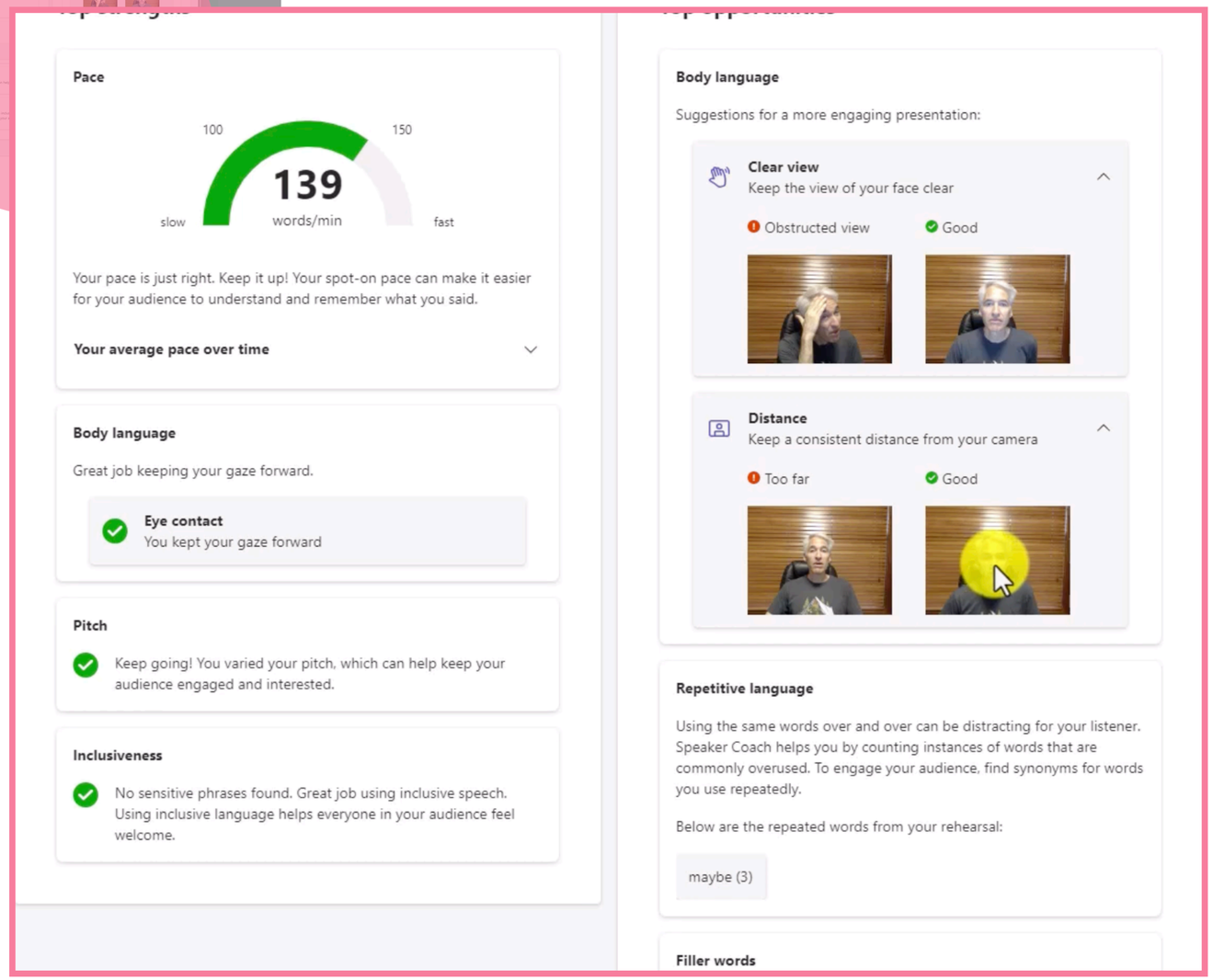
Visuals inside the Card

Show trending arrows? Add color?

Limitations: Microsoft cannot define what is **good versus bad trend** Ie. Can show up arrow as green or down as red

INSIGHTS DISCOVERABILITY FROM ACCELERATORS



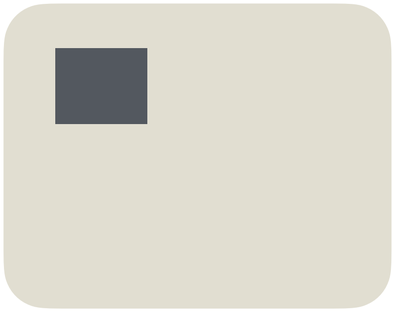


Establishing a framework

Auditing all insight cards across all learning accelerators.

Creating a design framework and establish rules for cohesion ie. How does the established styles in Reading Progress this look in Speaker Progress?

INSIGHTS DISCOVERABILITY FROM ACCELERATORS



Card
(single number)

Modal
(1 graph max)

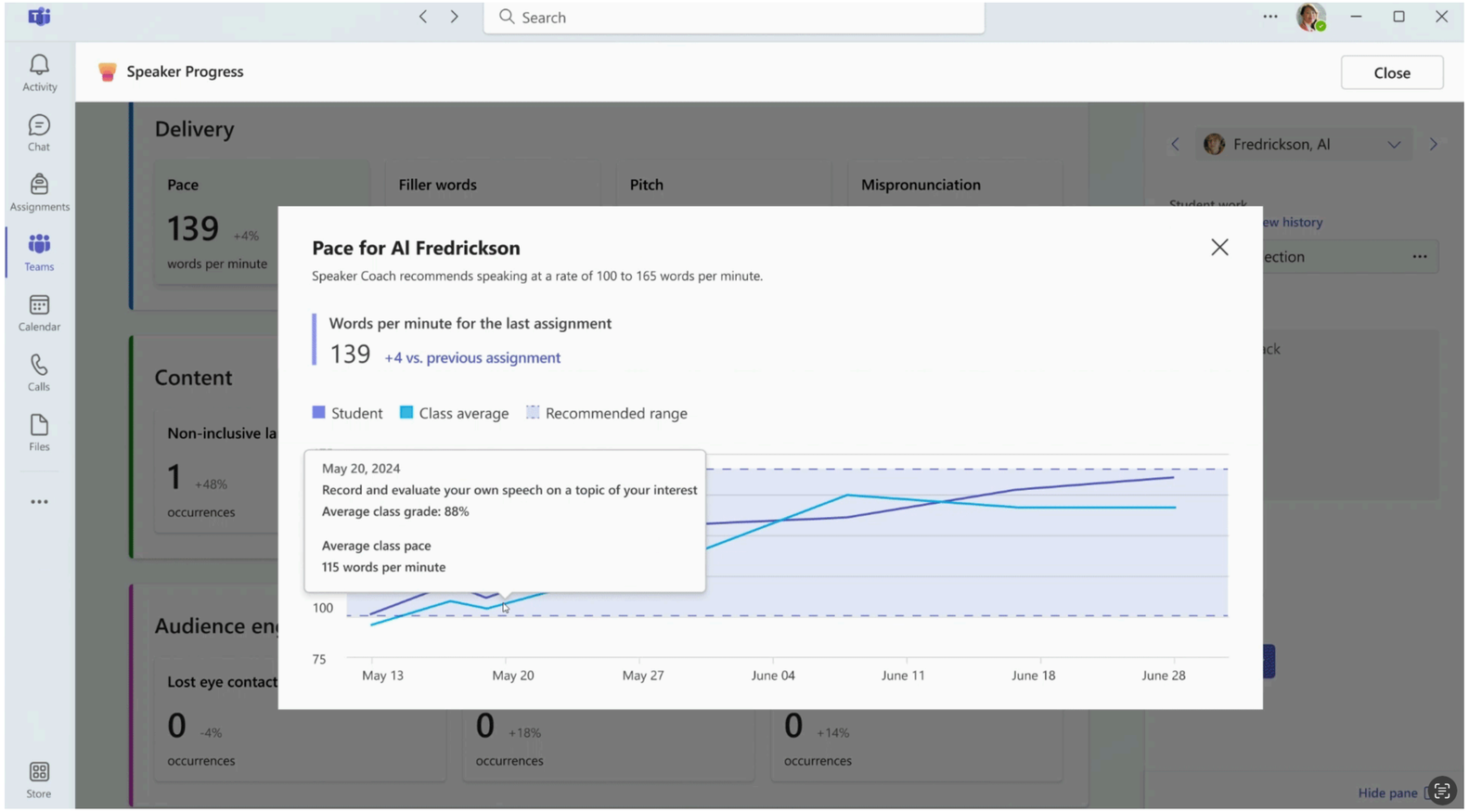
Insights
(Full historical data)

Progressive disclosure

Switching right from Assignments to **Insights.** Is disorienting, keeping in mind teachers are in the middle of grading usually when on review screen. Keep user flow focus

Progressive disclosure slows down the switch from Assignments to Insights - key part of the framework

INSIGHTS DISCOVERABILITY FROM ACCELERATORS



Progressive disclosure

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Progressive disclosure slows down the switch from Assignments to Insights - key part of the framework

INSIGHTS DISCOVERABILITY FROM ACCELERATORS

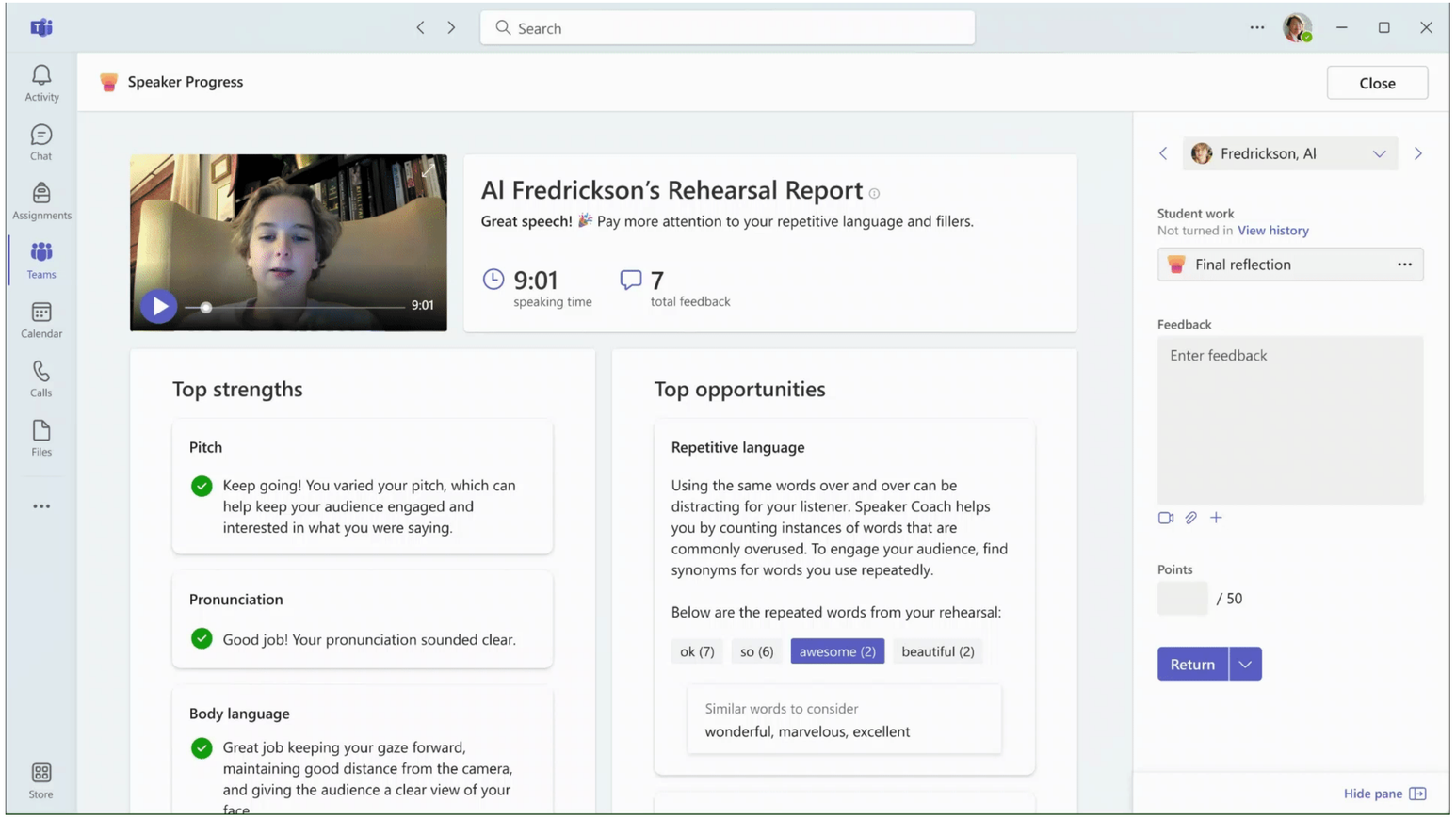
Learnings

I fell in love with my card hovering idea, building out a clickable prototype using real Fluent components that I didn't see a blindspot.

Students and teachers using iPads.

Rework interaction design without hover state

INSIGHTS DISCOVERABILITY FROM ACCELERATORS



Insights Cards Final

Framework used across all assignment types

Progressive Disclosure

No card flips. Click state on desktop translates to tap only on mobile/iPad

Increased card clicks and traffic to Insights by 40%

OTHER FRAMEWORK WORK FOR MICROSOFT EDU

How might we define a framework for Education specific AI copilot experiences horizontally?

COPILOT

How might we define a framework for Education specific AI copilot experiences horizontally?

Finding partners (and herding cats), setting up FigJam workshops, find alignment

COPILOT

How might we define a framework for Education specific AI copilot experiences horizontally?

Research

How do educators think of AI? What problems do they need the most help with? Trust and comfort level with AI?

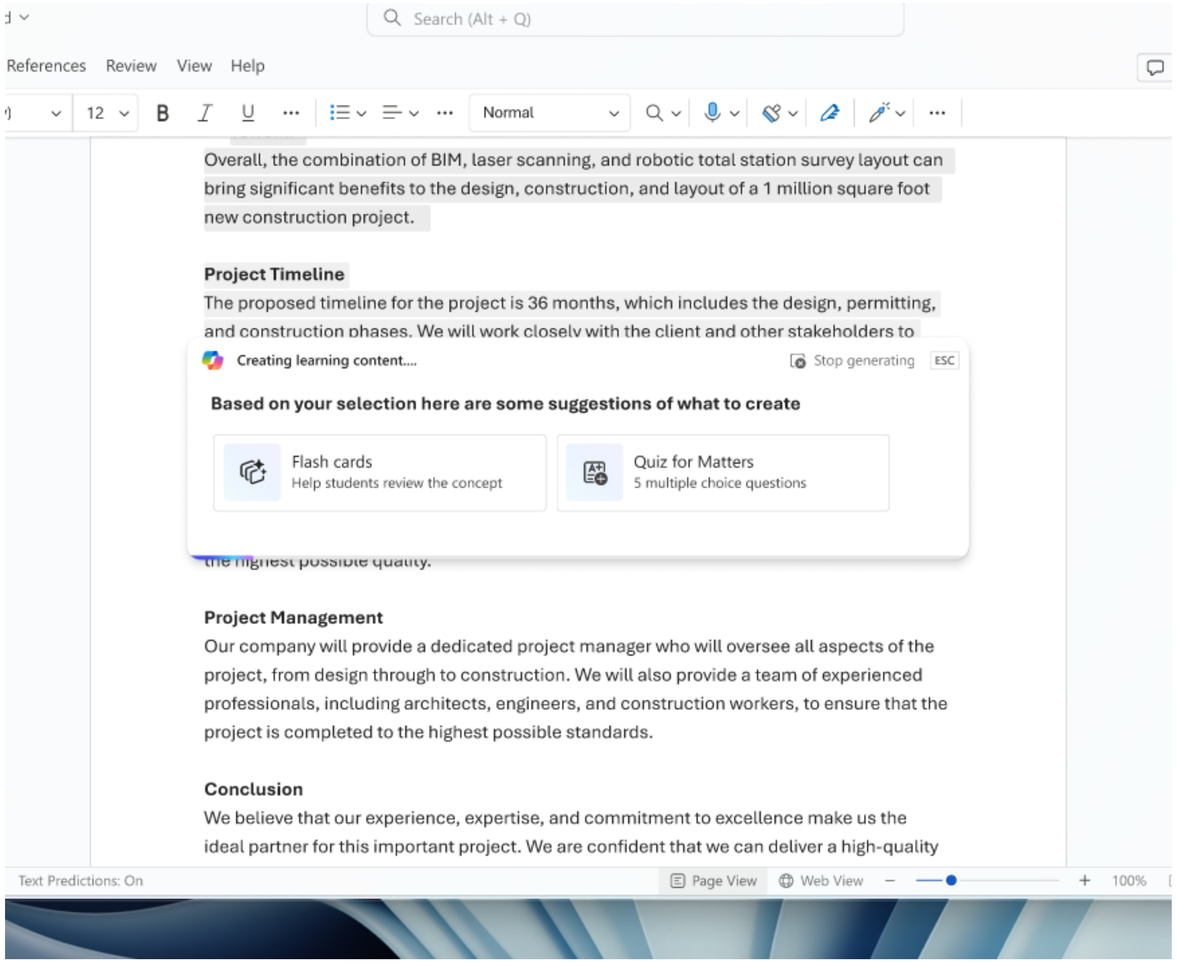
COPILOT

How might we define a framework for Education specific AI copilot experiences horizontally?

Outcomes

1. Experience framework when injecting Copilot into Education products (**chat pane and in context**)

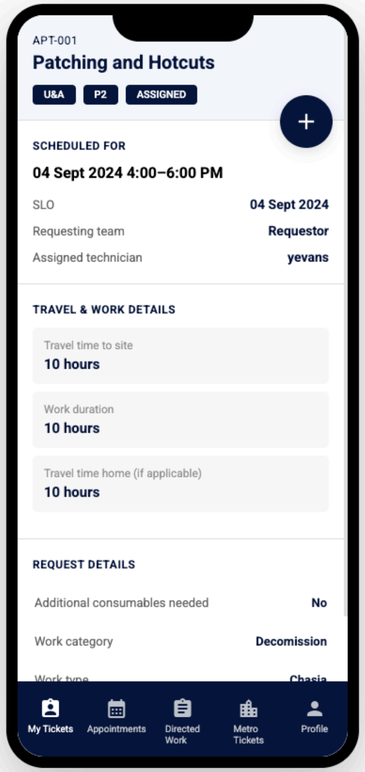
3. Informing AI first feature work: **identifying key feature opportunities** based on where teachers/students are today with their understanding and comfort with AI (ie. **AI Rubrics > AI grading)**



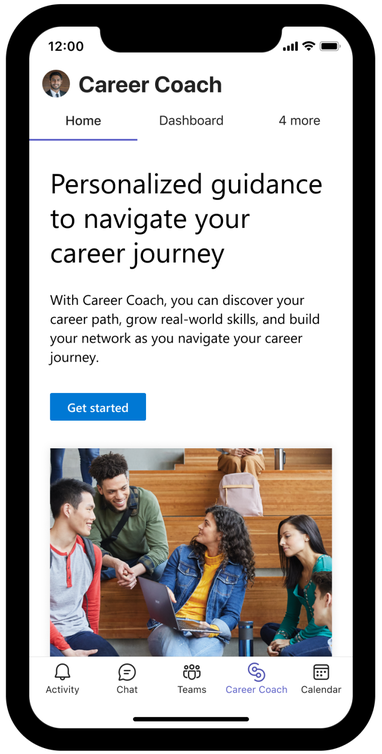
SUMMARY

Google NetOps

Moose



Metro Owner platform + field worker mobile app that made automation possible across **10** workstreams, **4** timezones, and **3** user groups. **30%** productivity gain.



Microsoft

Learning Accelerators

Frameworks for progressive disclosure of Insights and Education AI Copilot experiences

Increase Insights Discoverability by **40%**

THANK YOU! QUESTIONS?